

Non-Holonomic Gradient Play: Stability and Deception in Games on Manifolds

Mahmoud Abdelgalil, Miroslav Krstic, and Jorge I. Poveda

Abstract—We study generalized equilibrium seeking in multi-agent systems whose joint state evolves on a manifold and whose agents act through state-dependent, potentially non-holonomic control vector fields. Under a bundle-splitting condition, we show that the averaged interaction dynamics admit an intrinsic representation as a sum of projected Riemannian gradients, giving rise to a form of *non-holonomic gradient play*. We characterize the local stability of equilibria through an intrinsic linearization that explicitly captures the roles of the Riemannian connection and the non-holonomy of the actuation frame. The resulting framework recovers gradient-play dynamics in Euclidean spaces and finite-dimensional manifolds as special cases while accommodating non-holonomic actuation. We then develop a geometric theory of deception, in which a malicious agent exploits privileged information about other agents' learning rules to shift the equilibrium of the interaction. We show that deception effectively tilts the Riemannian geometry perceived by the victim agents, thereby distorting their projected gradient directions. We establish that exponential stability of the deception-free equilibrium persists over an open set of deception parameters and derive an explicit first-order characterization of the resulting equilibrium displacement. Analytical examples illustrate the proposed framework and its implications for equilibrium manipulation.

Index Terms—Multi-agent learning, games on manifolds, equilibrium seeking, deception.

I. INTRODUCTION

STRATEGIC interactions between autonomous decision-making agents are often constrained by geometry, with examples spanning many practical engineering problems, e.g., rigid body attitude coordination [1], [2], distributed resource allocation [3], [4], and synchronization of coupled oscillators [5]. These settings have two distinguishing structural features. First, the joint state of the interaction evolves on a manifold rather than on a vector space. Second, each agent can influence a restricted set of directions of the joint state, and those admissible directions typically vary with the state itself. That is, each agent acts through a family of state-dependent control vector fields. If, in addition, the agents pursue different goals described by payoff functions, the natural model of their

interaction is given by *gradient play dynamics* wherein each agent seeks to maximize its own payoff along the vector fields it controls [6], [7]. Therefore, game-theoretic notions of equilibrium are typically invoked to characterize the behavior of the interaction.

The earliest study of gradient play in games with continuous strategy spaces seems to be [8], [9], where differential characterizations of equilibria and of their local stability are given for games whose strategy spaces are finite-dimensional manifolds. However, the learning dynamics considered therein are expressed via partial derivatives of the payoffs with respect to a given choice of coordinates. Since gradient vector fields on a manifold are defined only relative to a Riemannian metric, and no such metric is introduced in [8], [9], neither the dynamics nor the ensuing stability conditions are coordinate independent. Moreover, in these works each agent actuates the coordinates of its own strategy space directly and independently, a framework that does not accommodate agents that act through a restricted family of vector fields, let alone a non-commuting one, even if the family collectively spans the tangent bundle of the manifold. Such structures arise naturally in nonholonomic and underactuated systems, including wheeled mobile robots, autonomous vehicles, robotic manipulators, and aerial and marine vehicles, where admissible motions are constrained by the system's actuation geometry. To the best of our knowledge, an intrinsic theory of gradient play in games whose joint state evolves on a manifold and whose agents are constrained to act through state-dependent, possibly non-commuting, vector fields, does not exist in the literature.

In parallel, there has been an active research thread on model-free equilibrium seeking [3], [10]–[13], which dispenses with explicit gradient information altogether and relies on zeroth-order information only, i.e., payoff measurements, to approximate gradient play dynamics. However, with the exception of specific synchronization problems [14]–[16], existing developments have largely been restricted to either Euclidean strategy spaces or single-agent settings [17], [18].

Motivated by the previous background, the first contribution of the present manuscript is precisely an *intrinsic theory of gradient play* for multi-agent systems evolving on smooth manifolds. Specifically, under a natural bundle-splitting assumption on the agents' actuation directions, we construct a Riemannian metric on the state space with respect to which the interaction dynamics amounts to a sum of *projected Riemannian gradients*, one per agent, with each projection being onto the distribution spanned by the corresponding agent's control vector fields. We refer to the resulting dynamics as

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non-holonomic gradient play, and to the corresponding vector field as the *non-holonomic pseudo-gradient*. This representation yields an equilibrium characterization that is coordinate independent. In particular, equilibria are determined solely by the payoffs of the agents and the geometry of the actuation directions. We then characterize the structural and the asymptotic stability properties of equilibria via an intrinsic linearization of the non-holonomic pseudo-gradient, which decomposes into two types of terms: Riemannian Hessian terms, and *Connection* terms. The latter type encodes the non-holonomy of the actuation frame and has no counterpart in the coordinate-based analysis of [8], [9]. Sufficient stability conditions of the block Gershgorin type are derived, and two special cases are worked out in detail. First, when each agent's distribution is involutive, we show that the diagonal blocks of the linearization reduce to intrinsic Hessians of restricted payoffs. When, in addition, the agents' vector fields mutually commute, the state space acquires a local product structure and the characterizations of [8], [9] are recovered as the flat special case. The identical interest game [19], wherein the dynamics reduces to an intrinsic gradient flow, is likewise subsumed.

The second contribution concerns *deception* under asymmetric information, and it is here that the generality of the first contribution proves instrumental. Specifically, in contested and adversarial environments, the assumption that all agents are equally informed is frequently violated. Indeed, an agent may be able to measure, in real time, the inputs applied by its competitors, or may possess privileged knowledge of their learning rules. Recent works [20]–[24] have shown that, on Euclidean product strategy spaces with additive actuation structures, such asymmetry enables a malicious agent with access to privileged information to relocate the equilibrium of the interaction by manipulating only its own inputs, without tampering with other agents' measurements or actuation. The mechanisms of [20]–[22], however, are tailored to the additive structure of $\mathbb{R}^n$ and does not extend to the geometric setting considered here. In contrast, within the present framework, we show that deception admits an intrinsic geometric description: *by embedding the victim's exploration signals into its own inputs, a deceiver tilts the victim's effective actuation bundle along its own, and, thereby, distorts the Riemannian projections of the victim in the nonholonomic gradient*. As a result, the tilted bundles generically fail to be involutive even when the nominal ones are, and, therefore, analyzing deception necessitates the general theory we develop herein. In particular, we prove that exponentially stable equilibria persist, together with their stability, over an open set of deception parameters, and we derive an explicit first-order formula for the equilibrium shift as a function of those parameters.

The theory developed herein is formulated directly in terms of the non-holonomic gradient-play dynamics and is therefore independent of the particular mechanism through which such dynamics are generated in the multi-agent system. In particular, the framework applies whenever the target behavior of a learning or control algorithm takes the form of non-holonomic gradient play. One important instance of this setting arises in model-free equilibrium seeking via Lie-bracket averaging. As discussed in the sequel, the interaction dynamics studied

herein emerge as the Lie-bracket approximation [25]–[27] of a control-affine system driven by payoff-only feedback laws. In this case, the proposed theory characterizes the target behavior of model-free equilibrium-seeking schemes along the lines of [3], [10]–[13], [15], but for games evolving on manifolds and subject to actuation constraints. More generally, the same target dynamics can also arise through other approximation mechanisms, including stochastic approximations [28]. Thus, Lie-bracket-based equilibrium-seeking dynamics provide one concrete motivation for the class of systems studied herein, but the results apply more broadly to any learning or control architecture whose effective dynamics are described by non-holonomic gradient play.

The remainder of the manuscript is organized as follows. Section II collects notation and geometric preliminaries. Section III presents the interaction model, its averaged dynamics, and the information-asymmetry model. Section IV develops the intrinsic theory of non-holonomic gradient play: the constructed metric, the equilibrium characterization, the intrinsic linearization, and the involutive and commuting special cases, illustrated throughout by two running examples on the sphere and on the simplex. Section V develops the theory of deception: the tilting mechanism, the persistence of stable equilibria, and the first-order sensitivity of the equilibrium map. Section VI collects the proofs of the main results, and Section VII concludes with directions for future work.

II. PRELIMINARIES AND NOTATION

Throughout, we use standard differential geometric notation consistent with, e.g., [29]. Let $\mathcal{M}$ be a smooth n -dimensional manifold, $T\mathcal{M}$ be its tangent bundle, $T^*\mathcal{M}$ its cotangent bundle. We use $\Gamma(T\mathcal{M} \otimes \cdots \otimes T\mathcal{M} \otimes T^*\mathcal{M} \otimes \cdots \otimes T^*\mathcal{M})$ to denote the space of smooth sections of the bundle $T\mathcal{M} \otimes \cdots \otimes T\mathcal{M} \otimes T^*\mathcal{M} \otimes \cdots \otimes T^*\mathcal{M}$, so that the space of smooth vector fields is $\Gamma(T\mathcal{M})$, the space of smooth differential 1-forms is $\Gamma(T^*\mathcal{M})$, and so on. For brevity, we also use $\mathfrak{X}(\mathcal{M})$ to denote the space of smooth vector fields on $\mathcal{M}$, and $\Omega(\mathcal{M})$ to denote the space of smooth differential 1-forms on $\mathcal{M}$. The symbol $\otimes$ denotes *tensor product* so that, if $f_1, f_2 \in \mathfrak{X}(\mathcal{M})$ and $\omega_1, \omega_2 \in \Omega(\mathcal{M})$, then $f_1 \otimes f_2 \in \Gamma(T\mathcal{M} \otimes T\mathcal{M})$, $\omega_1 \otimes \omega_2 \in \Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$, $\omega_i \otimes f_i \in \Gamma(T^*\mathcal{M} \otimes T\mathcal{M})$, and so on. For $f \in \mathfrak{X}(\mathcal{M})$ and J a smooth function, the Lie derivative of J along f is

$$L_f J(x) := dJ(x)[f(x)],$$

where $dJ \in \Omega(\mathcal{M})$ denotes the differential of J . A Riemannian metric on $\mathcal{M}$, which is a symmetric positive definite section of $\Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$, is denoted by g . Given a smooth function J , the Riemannian gradient with respect to g , denoted by $\text{grad}(J)$, is the unique vector field satisfying

$$g(\text{grad}(J), X) = dJ[X], \quad \forall X \in \mathfrak{X}(\mathcal{M}).$$

The Riemannian *Hessian* of J is the unique symmetric section of $\Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$ that satisfies

$$\text{Hess}(J)[X, Y] := g(\nabla_X \text{grad}(J), Y),$$

where $\nabla : \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$ is the Levi-Civita connection associated with the metric g . The fact that the

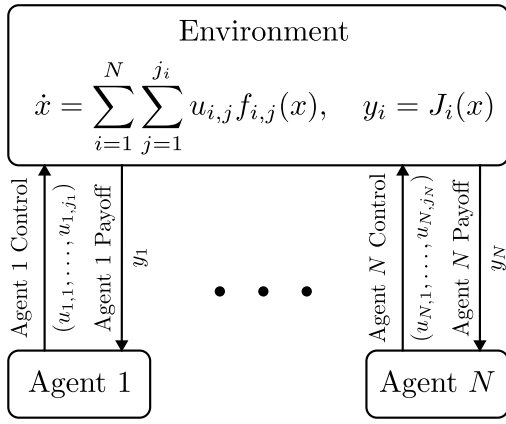


Fig. 1. Block diagram of the strategic interaction between N independent decision-making agents coupled only through the environment.

Riemannian Hessian is symmetric, i.e., $\text{Hess}(J)[X, Y] = \text{Hess}(J)[Y, X]$, follows from the torsion-free nature of the Levi-Civita connection. If $\mathcal{E} \subset T\mathcal{M}$ is a smooth sub-bundle and $P_{\mathcal{E}} : T\mathcal{M} \rightarrow \mathcal{E}$ is the g -orthogonal projection, then we define the *projected gradient*

$$\text{grad}_{\mathcal{E}}(J) := P_{\mathcal{E}} \text{grad}(J).$$

Finally, we use $\cdot|_x$ to denote the evaluation operation, i.e., if $f \in \mathfrak{X}(\mathcal{M})$, then $f|_x = f(x) \in T_x\mathcal{M}$, and so on.

III. PROBLEM FORMULATION

In this section, we describe the problem of interest of this paper, as well as its connection to the standard (generalized) equilibrium-seeking problems.

A. Geometric Equilibrium-Seeking in Games

We consider N independent decision-making agents interacting through a strategic environment modeled as a smooth n -dimensional manifold $\mathcal{M}$. The i^{th} agent influences the environmental state $x \in \mathcal{M}$ through a collection of vector fields $\{f_{i,j}\}_{j=1}^{d_i} \subset \mathfrak{X}(\mathcal{M})$, with its actions determined by the control inputs $(u_{i,1}, \dots, u_{i,d_i}) \in \mathbb{R}^{d_i}$. These inputs act on the environment through a control-affine structure, yielding the dynamics

$$\dot{x} = \sum_{i=1}^N \sum_{j=1}^{d_i} u_{i,j} f_{i,j}(x), \quad x \in \mathcal{M}. \quad (1)$$

In the non-cooperative setting, each agent is interested in optimizing their own payoff, which may, in general, be different from other agents' payoffs. We model this setting by assigning to each agent a sufficiently regular functions J_i , for $i \in \{1, 2, \dots, N\}$. A block-diagram representation of the class of systems considered in this paper is shown in Figure 1. In multi-agent systems evolving in $\mathbb{R}^n$, it is common to consider decentralized feedback laws u_i based on the partial gradient (i.e., pseudogradient) information of the cost functions of the agents, i.e., gradient play [6], [7]. For the control-affine structures of the form (1), the natural extension is given by the feedback law:

$$u_{ij}(x) = \kappa_i L_{f_{i,j}}(J_i)(x), \quad (2)$$

where $\kappa_i > 0$ is a tunable gain. This choice leads to the following dynamics:

$$\dot{x} = \sum_{i=1}^N \kappa_i \sum_{j=1}^{d_i} L_{f_{i,j}}(J_i)(x) f_{i,j}(x), \quad x \in \mathcal{M}. \quad (3)$$

These dynamics constitute the main object of study of this paper. In particular, as shown in the next section, under mild regularity conditions on $\mathcal{M}$, $\{f_{i,j}\}$, and J_i , they admit an intrinsic interpretation as a non-holonomic gradient play, encompassing standard gradient-play dynamics as a particular case.

Remark 1 (Model-Free Implementation): Systems of the form (3) can also emerge in the analysis of model-free decision-making algorithms that only have access to measurements of J_i via Lie-bracket averaging [23], [25], [30]. In this case, the inputs of the i^{th} -agent in (1) may be taken as the following oscillatory feedback law:

$$u_{i,j}(x, t) = \sqrt{4\pi\kappa_i\omega_{i,j}} \cos(2\pi\omega_{i,j}\omega t) F_{i,j}^1(J_i(x)) - \sqrt{4\pi\kappa_i\omega_{i,j}} \sin(2\pi\omega_{i,j}\omega t) F_{i,j}^2(J_i(x)), \quad (4)$$

where $\omega \in \mathbb{R}_{>0}$ is a tunable parameter, the constants $\cup_{i=1}^N \{\omega_{i,j}\}_{j=1}^{d_i} \subset \mathbb{Q}_{>0}$ are any collection of positive rational numbers such that $\omega_{i,j} = \omega_{i',j'}$ if and only if $i = i'$ and $j = j'$, the constants $\kappa_i > 0$ are positive gains, and $F_{i,j}^k : \mathbb{R} \rightarrow \mathbb{R}$ is any collection of sufficiently regular functions that satisfy

$$F_{i,j}^2(y) = -F_{i,j}^1(y) \int F_{i,j}^1(y)^{-2} dy. \quad (5)$$

An example of such functions is $F_{i,j}^1(y) := \cos(y)$ and $F_{i,j}^2(y) := -\sin(y)$, originally introduced in [30]; see also [26], [31]. Regardless of the particular choice of these functions, the control-affine system (1) under the class of output-feedback laws (4) is known to approximate, as $\omega \rightarrow \infty$, an associated *Lie-Bracket Approximation* (LieBrA) system [25], [27], which in the present setting coincides precisely with (3). In particular, the stability properties of (3) are inherited, in a *practical* sense, by the closed-loop system (1)–(4); see [25], [27], [32]. This approximation therefore allows the behavior of the model-free (i.e., payoff-based) dynamics (1)–(4) to be characterized through the model-based (i.e., derivative-based) target dynamics (3). $\square$

Remark 2: When $\mathcal{M} = \mathbb{R}^n$ and the agents' vector fields are decoupled, the analogue of system (3) is the standard *gradient-play dynamics* $\dot{x}_i = \nabla_i J_i$. While the convergence and stability properties of standard gradient play are well understood, the corresponding properties of the more general manifold-constrained dynamics (3), particularly in the presence of state-dependent and potentially non-commuting actuation directions, have remained largely unexplored in the context of games. $\square$

B. Asymmetric Information and Deception

The model presented thus far is formulated under the assumption of *information symmetry*, i.e., that none of the decision-making agents has access to *privileged* information concerning the feedback laws implemented by other agents.

However, as shown recently in [20]–[22], when this assumption is violated, i.e., in the presence of *information asymmetry*, agents with access to privileged information can manipulate the asymptotic behavior of the multi-agent system through careful manipulation of their own control inputs only. In particular, such *deceptive* agents can systematically *bias* the steady-state behavior of system (1) without ever interfering with the control actions of the *oblivious agents* or their measurements. For recent related tutorials on *deception* in game theory, we refer the readers to [24], [33].

The context in which the phenomenon of *deception* has been studied in the literature of Nash-Seeking is that of repeated static non-cooperative games on *flat spaces*, particularly, in $\mathbb{R}^n$. However, deception in generalized equilibrium-seeking problems on smooth manifolds remains largely unexplored, particularly *with respect to its interplay with the underlying geometry*. Our goal in this paper is to extend the study of deception in adaptive multi-agent decision-making to the setting described by (1).

Remark 3: To illustrate how asymmetric information can be leveraged to bias the asymptotic behavior of the target system (3), we consider the setting of Lie-bracket averaging discussed¹ in Remark 1, and for simplicity, a two-agent setting; the extension to $N > 2$ agents is discussed in Remark 10. Suppose that the 1st agent, referred to as the *deceptive agent*, has access to the control input (4) generated by the 2nd agent, referred to as the *oblivious agent*. By leveraging this privileged information, the deceptive agent modifies its feedback law according to

$$\tilde{u}_{1,j}(x, t) := u_{1,j}(x, t) + \sum_{k=1}^{d_2} \gamma_k^j u_{2,k}(x, t), \quad (6)$$

where $\{\gamma_k^j\}$ is a collection of tunable deception parameters selected by the deceptive agent. Under this modification, the resulting closed-loop system can be equivalently written as

$$\dot{x} = \sum_{i=1}^2 \sum_{j=1}^{d_i} u_{i,j}(x, t) \tilde{f}_{i,j}(x), \quad (7a)$$

where

$$\tilde{f}_{1,j} := f_{1,j}, \quad \tilde{f}_{2,j} := f_{2,j} + \sum_{k=1}^{d_1} \gamma_j^k f_{1,k}. \quad (7b)$$

Consequently, the associated Lie-Bracket Approximation target system now takes the form

$$\dot{x} = \sum_{i=1}^2 \kappa_i \sum_{j=1}^{d_i} L_{\tilde{f}_{i,j}}(J_i)(x) \tilde{f}_{i,j}(x). \quad (8)$$

Thus, the deceptive interaction preserves the structural form of (3), but effectively *tilts* the vector fields associated with the oblivious agent according to (7b). In this sense, the deceptive agent uses its privileged knowledge of the oblivious

agent's control law to reshape the effective directions along which the latter's payoff influences the target dynamics. As a consequence, the equilibria of (8) will, in general, differ from those of the nominal system (3), with the nominal dynamics recovered when $\gamma_j^k \equiv 0$. A formal characterization of the resulting deceptive dynamics and their asymptotic behavior is presented in Section V. $\square$

We conclude the current section with a discussion on the assumption of asymmetric information adopted in (6). To that end, it is worth mentioning that the particular form of information asymmetry adopted in [20]–[22], [33] is *tailored* for the classical Nash-Seeking algorithms that evolve in $\mathbb{R}^n$. Consequently, these assumptions cannot be directly applied to the model (1), as they fundamentally rely on the *additive structure* of Euclidean spaces. Moreover, the type of interactions studied in [20]–[22], [33] necessitates that the vector fields $f_{i,j}$ are constant and completely decoupled, i.e., that $\mathcal{M} = \mathbb{R}^{j_1} \times \dots \times \mathbb{R}^{j_N} \simeq \mathbb{R}^{j_1 + \dots + j_N}$, and $f_{i,j}(x) = \partial_{(i-1)d_i + j}$. Herein, we relax both of these assumptions as evidenced by the general model (1), and, as a result, a stronger notion of information asymmetry is required to obtain a suitable generalization of the deception mechanism in [20]–[22], [33]. Possible relaxations of this assumption will be pursued in future work.

IV. NON-HOLONOMIC GRADIENT PLAY DYNAMICS

In this section, we focus our attention on system (3), repeated below for convenience,

$$\dot{x} = \sum_{i=1}^N \kappa_i \sum_{j=1}^{d_i} L_{f_{i,j}}(J_i)(x) f_{i,j}(x). \quad (3)$$

Characterizing the asymptotic behavior of (3) necessitates identifying its equilibria as the first step. However, without further assumptions on the vector fields $f_{i,j}$, finding the equilibria of (3) requires solving the general system of equations

$$\sum_{i=1}^N \kappa_i \sum_{j=1}^{d_i} L_{f_{i,j}}(J_i)(x) f_{i,j}(x) = 0. \quad (9)$$

An issue with (9) is the fact that, in general, the equilibria of (9) will depend on the choice of the gains κ_i , a choice that is arbitrary and independent of the functions J_i . Nevertheless, there are natural assumptions that allow us to characterize those equilibria that satisfy (9) for any choice of the gains κ_i . To that end, we impose the following assumption.

Assumption 1: There exists an open sub-manifold $\mathcal{U} \subset \mathcal{M}$ such that, for all $x \in \mathcal{U}$, we have

$$T_x \mathcal{U} = \mathcal{E}_1(x) \oplus \dots \oplus \mathcal{E}_N(x), \quad (10)$$

where $\mathcal{E}_i$ is a *regular* distribution on $\mathcal{U}$ defined by the span of the i^{th} -family of vector fields $\{f_{i,j}\}_{j=1}^{d_i}$, i.e., $\mathcal{E}_i(x) := \text{span}_{\mathbb{R}} \{f_{i,j}(x)\}_{j=1}^{d_i}$. $\square$

In words, Assumption 1 asks that the family of vector fields $\bigcup_{i=1}^N \{f_{i,j}\}_{j=1}^{d_i}$ spans the tangent space of the manifold $\mathcal{M}$ for all points in an open set $\mathcal{U}$ and that the intersection of the spans of any two distinct families $\{f_{i,j}\}_{j=1}^{d_i}$ and $\{f_{i',j}\}_{j=1}^{d_{i'}}$ is trivial

¹Again, the particular approximation mechanism is immaterial to results of this paper, as the target system (3) can arise from different classes of learning algorithms, including those based on Lie-bracket averaging, stochastic approximations, etc.

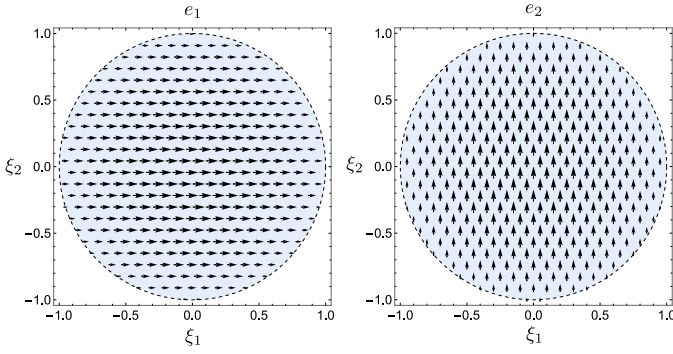


Fig. 2. An illustration of the vector fields e_i for Example 1 in the ξ -coordinates. The arrows' sizes represent the magnitude of the vector field.

whenever $i \neq i'$. In the context of the multi-agent strategic interaction modeled by (1), Assumption 1 essentially requires that the control vector fields influenced by the i^{th} -agent are linearly independent from the span of the control vector fields influenced by every other agent at every environment state $x \in \mathcal{U}$ and that, collectively, the family $\cup_{i=1}^N \{f_{i,j}\}_{j=1}^{d_i}$ spans the entire tangent bundle $T\mathcal{U}$. We now illustrate this assumption with some examples. In these examples, the distributions $\mathcal{E}_i$ are all rank 1, i.e., $d_i = 1$ for all $i \in \{1, \dots, N\}$. To simplify notation, we use $e_i := f_{i,1}$ to get rid of superfluous indices.

Example 1: We consider a two-player decision-making problem on the sphere $\mathcal{M} := \mathbb{S}^2 = \{x \in \mathbb{R}^3 \mid x^\top x = 1\} \subset \mathbb{R}^3$, which is a 2-dimensional Euclidean sub-manifold. Let the vector fields e_i be defined in the ambient coordinates $x = (x_1, x_2, x_3)$ by

$$e_i(x) := x_3 \partial_i - x_i \partial_3,$$

which can be shown to be tangent to $\mathbb{S}^2$. To verify Assumption 1 on a suitable submanifold $\mathcal{U} \subset \mathcal{M}$, we define the function $\Delta(x) := \langle x, e_1(x) \times e_2(x) \rangle_{\mathbb{R}^3}$, where $\times$ is the cross product of vectors and $\langle \cdot, \cdot \rangle_{\mathbb{R}^3}$ is the inner product between vectors in $\mathbb{R}^3$. Direct computation gives that

$$\Delta(x) = x_3(x_1^2 + x_2^2 + x_3^2) = x_3.$$

Therefore, since the cross product of a pair of vectors in $\mathbb{R}^3$ vanishes if and only if they are linearly dependent, it follows that the vector fields e_i are linearly independent and span the tangent space of $\mathbb{S}^2$ everywhere except at the degenerate set $\{x \in \mathbb{S}^2 \mid x_3 = 0\}$. In particular, by restricting our analysis to the open sub-manifold

$$\mathcal{U} := \{x \in \mathbb{S}^2 \mid x_3 > 0\},$$

we are guaranteed that, if $\mathcal{E}_i(x) := \text{span}\{e_i(x)\}$, then Assumption 1 holds on $\mathcal{U}$. Moreover, the submanifold $\mathcal{U}$ is diffeomorphic to the open disk $\mathbb{D}^2 := \{\xi = (\xi_1, \xi_2) \in \mathbb{R}^2 \mid \xi_1^2 + \xi_2^2 < 1\}$ via the diffeomorphism $(\xi_1, \xi_2) = \varphi(x) = (x_1, x_2)$, which will be convenient for visualizations. For instance, the vector fields e_i are depicted in Figure 2 using the ξ -coordinates. It is also worth mentioning that the vector fields e_i do not commute. Indeed, direct computation gives

$$[e_1, e_2](x) = x_2 \partial_1 - x_1 \partial_2,$$

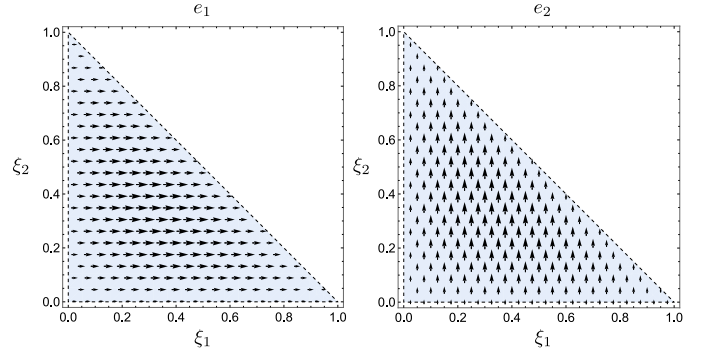


Fig. 3. An illustration of the vector fields e_i for Example 2 in the ξ -coordinates. The arrows' sizes represent the magnitude of the vector field.

which, generically, is non-vanishing on $\mathcal{U}$. $\square$

Example 2: Consider a two-player resource allocation problem on the 2-simplex $\mathcal{M} = \mathcal{S}^2 := \{x \in \mathbb{R}^3 \mid x_i \geq 0, \sum_{i=1}^3 x_i = 1\}$, which is a 2-dimensional Euclidean sub-manifold. Consider the vector fields e_i defined in the ambient coordinates as follows:

$$e_i(x) := x_1 x_2 x_3 (\partial_i - \partial_3).$$

Direct computation shows that the vector fields e_i are tangent to $\mathcal{M}$ and span the tangent space everywhere except at the degenerate set $\{x \in \mathcal{M} \mid x_1 x_2 x_3 = 0\}$, which corresponds to the topological boundary of $\mathcal{M}$, denoted by $\partial\mathcal{M}$. Hence, by restricting our analysis to the open submanifold $\mathcal{U} := \mathcal{M} \setminus \partial\mathcal{M}$, we are guaranteed that, with $\mathcal{E}_i = \text{span}\{e_i\}$, then Assumption 1 holds on $\mathcal{U}$. Moreover, similar to the previous example, the submanifold $\mathcal{U}$ is diffeomorphic to an open subset of the plane via the diffeomorphism $(\xi_1, \xi_2) = \varphi(x) = (x_1, x_2)$, which will be convenient for visualizations. Indeed, the vector fields e_i are depicted in Figure 3. We also point out that the vector fields e_i do not commute. In fact, direct computation gives

$$[e_1, e_2](x) = x_1 x_2 x_3 (x_1(x_3 - x_2)\partial_1 + x_2(x_1 - x_3)\partial_2 + x_3(x_2 - x_1)\partial_3),$$

which, generically, is non-vanishing on $\mathcal{U}$. $\square$

We will be mainly interested in the behavior of the dynamical system (3) within the open sub-manifold $\mathcal{U}$. As such, Assumption 1 allows us to endow system (3) with a natural geometric interpretation. In particular, our first result shows that *the flow of (3) is a Riemannian pseudo-gradient flow on $\mathcal{M}$* . All the proofs are presented in Section VI.

Theorem 1: Let Assumption 1 be satisfied. Then, there exists a Riemannian metric g on the sub-manifold $\mathcal{U} \subset \mathcal{M}$ such that system (3) takes the form

$$\dot{x} = \sum_{i=1}^N \kappa_i \text{grad}_{\mathcal{E}_i}(J_i)(x), \quad (11)$$

where $\text{grad}_{\mathcal{E}_i}(J_i)$ is the vector field obtained through orthogonal projection of $\text{grad}(J_i)$ with respect to g onto $\mathcal{E}_i$. $\square$

In view of (11), we refer to the Lie-bracket approximation system (3) as a *non-holonomic gradient play*: each agent

ascends its own payoff, but only along the state-dependent, and generically non-commuting, directions available to it. In the special case $\mathcal{M} = \mathbb{R}^n$ with $f_{i,j} = \partial_{(i,j)}$ in the standard coordinates, system (11) reduces to the classical gradient play dynamics studied in [8], [9].

Remark 4: The proof of Theorem 1 is constructive in the sense that it explicitly defines the Riemannian metric on the sub-manifold $\mathcal{U}$ for which the conclusions of the Theorem hold. Moreover, if the sub-manifold $\mathcal{U}$ contains any equilibria of the system (3), then, thanks to Assumption 1, the condition (9) simplifies to the natural geometric condition

$$\mathbf{grad}_{\mathcal{E}_i}(J_i)|_{x^*} = 0, \quad i \in \{1, \dots, N\}. \quad (12)$$

Note that, unlike (9), the condition (12) is independent of the gains κ_i , i.e., no agent can change the equilibria of (3) in the sub-manifold $\mathcal{U}$ by amplifying their gain κ_i , which implies that such equilibria are solely determined by the payoffs J_i and the family of vector fields $\cup_{i=1}^N \{f_{i,j}\}_{j=1}^{d_i}$, i.e., by the properties of the environment (cf. Figure 1). Therefore, we may take $\kappa_i = 1$ for all i , which is equivalent to absorbing the constants κ_i into the payoff functions, without affecting the equilibria of (3) in the sub-manifold $\mathcal{U}$. $\square$

Having characterized the equilibria of the interaction dynamics (3), we now proceed to study their nature. Specifically, we would like to characterize sufficient conditions under which an equilibrium point, i.e., a point satisfying (12), is *isolated* and locally exponentially stable. This is achieved using the *intrinsic Jacobian*.

The intrinsic Jacobian of the vector field defining (3) is the $(1,1)$ -tensor $A : \mathfrak{X}(\mathcal{U}) \rightarrow \mathfrak{X}(\mathcal{U})$ defined by

$$A[X] := \nabla_X \left(\sum_{i=1}^N \mathbf{grad}_{\mathcal{E}_i}(J_i) \right), \quad (13)$$

for any $X \in \mathfrak{X}(\mathcal{U})$, where we utilize the linearity of the Levi-Civita connection. The local properties of an equilibrium point $x^* \in \mathcal{U}$ satisfying (12) is dictated by the *eigenvalues* of the linear map $A|_{x^*} : T_{x^*}\mathcal{U} \ni v \rightarrow A|_{x^*}[v] \in T_{x^*}\mathcal{U}$, i.e. the restriction of the Jacobian defined in (13) to the tangent space at the point x^* . In particular, we have the following proposition.²

Proposition 1: Let Assumption 1 be satisfied for some $\mathcal{U} \subset \mathcal{M}$ and let $x^* \in \mathcal{U}$ be such that (12) holds. Then, the following statements are true:

- 1) The point x^* is an isolated equilibrium for system (3) if the map $A|_{x^*}$ is an automorphism of $T_{x^*}\mathcal{U}$.
- 2) The point x^* is locally exponentially stable for system (3) if and only if the map $A|_{x^*}$ is Hurwitz. $\square$

Example 3 (Identical Interest Games): Before we proceed to analyze the structure of the Jacobian (13), it is illuminating to consider the case where $J_1 = J_2 = \dots = J_N = J$, i.e., the *identical interest* game in which all agents share the same payoff [19]. In that case, system (3) is, essentially, an intrinsic

gradient flow³. In addition, the equilibrium condition (12) and the operator (13) simplify to

$$\mathbf{grad}(J)|_{x^*} = 0, \quad A[X] = \nabla_X \mathbf{grad}(J),$$

where, since $\{P_{\mathcal{E}_i}\}_{i=1}^N$ are a complete set of projection operators, we used the property that $\sum_{i=1}^N P_{\mathcal{E}_i} = I$. Applying Proposition 1 to this special case, we recover the standard conditions of optimization in the geometric setting. In particular, item 2) in Proposition 1 is equivalent to the condition that the Hessian is negative definite at x^* . $\square$

Given the discussion in Example 3, it is tempting to extend the intuition provided by the identical interest case to the general case. However, as will be shown in the sequel, the curvature induced by the geometry of the bundles $\mathcal{E}_i$ contributes in a non-trivial way to the Jacobian (13). Indeed, recalling that $\mathbf{grad}_{\mathcal{E}_i}(J_i) := P_{\mathcal{E}_i}[\mathbf{grad}(J_i)]$, where the $(1,1)$ -tensor $P_{\mathcal{E}_i}$ is the orthogonal projection operator with respect to the metric onto the distribution $\mathcal{E}_i$, and utilizing the product rule of the Levi-Civita connection, we see that

$$A = \sum_{i=1}^N P_{\mathcal{E}_i}[\nabla \mathbf{grad}(J_i)] + (\nabla P_{\mathcal{E}_i})[\mathbf{grad}(J_i)]. \quad (14)$$

From (14), it is immediately clear that the curvature of the bundles $\mathcal{E}_i$ has a non-negligible contribution to the Jacobian. Indeed, the second term in (14) does not vanish generically at an equilibrium point x^* since condition (12) may hold at a point $x^* \in \mathcal{U}$ even if $\mathbf{grad}(J_i)|_{x^*} \neq 0$. Using the orthogonal splitting (10), we may decompose A into a *block* form by defining the linear maps

$$A_{ij}[\cdot] = \sum_{k=1}^N P_{\mathcal{E}_i} \left[P_{\mathcal{E}_k}[\nabla_{P_{\mathcal{E}_j}[\cdot]} \mathbf{grad}(J_k)] + (\nabla_{P_{\mathcal{E}_j}[\cdot]} P_{\mathcal{E}_k})[\mathbf{grad}(J_k)] \right].$$

Since $\mathcal{E}_i$ are orthogonal, it follows that $P_{\mathcal{E}_i}P_{\mathcal{E}_k} = \delta_{ik}P_{\mathcal{E}_k}$, and, therefore, the tensor fields A_{ij} simplify to

$$A_{ij}[\cdot] = P_{\mathcal{E}_i}[\nabla_{P_{\mathcal{E}_j}[\cdot]} \mathbf{grad}(J_i)] + C_{ij}[\cdot], \quad (15a)$$

where C_{ij} are the tensor fields defined by

$$C_{ij}[\cdot] := \sum_{k=1}^N P_{\mathcal{E}_i}[(\nabla_{P_{\mathcal{E}_j}[\cdot]} P_{\mathcal{E}_k})[\mathbf{grad}(J_k)]]. \quad (15b)$$

From this block form, it is possible to derive a sufficient condition to ensure that the real parts of the eigenvalues of the linear map $A|_{x^*}$ are strictly negative and, therefore, that x^* is both isolated and locally exponentially stable. The following Proposition formalizes this sufficient condition:

Proposition 2: Let $x^* \in \mathcal{U}$ satisfy (12) and let A_{ij} be the blocks defined in (15). For each pair (i, j) , define $\|A_{ij}\|_g$ as the operator norm induced by g , i.e.,

$$\|A_{ij}\|_g := \sup_{v \in \mathcal{E}_j(x^*) \setminus \{0\}} \frac{\|A_{ij}v\|_g}{\|v\|_g},$$

³An intrinsic gradient flow is a gradient-based dynamical system defined directly on a manifold using its geometric structure, without relying on coordinates or an embedding into a Euclidean space.

²The proof is classical and therefore it is omitted.

and, for a linear map M on $\mathcal{E}_i(x^*)$, let $\sigma_{\min}^g(M) := \inf_{v \in \mathcal{E}_i(x^*) \setminus \{0\}} \|Mv\|_g / \|v\|_g$. Suppose that there exist positive constants $\chi_i > 0$ ($i = 1, \dots, N$) such that, for each i ,

$$\inf_{\Re(s) \geq 0} \sigma_{\min}^g(A_{ii}|_{x^*} - sI_i) > \sum_{j \neq i} \|A_{ij}|_{x^*}\|_g \frac{\chi_j}{\chi_i}, \quad (16)$$

where I_i denotes the identity map on $\mathcal{E}_i(x^*)$. Then $A|_{x^*}$ is an automorphism and Hurwitz. $\square$

In the remainder of this section, we impose the following assumption:

Assumption 2: For each $i \in \{1, \dots, N\}$, the family $\{f_{i,j}\}_{j=1}^{d_i}$ is linearly independent at every $x \in \mathcal{U}$. $\square$

Under this hypothesis, the Riemannian metric g on $\mathcal{U}$ constructed in the proof of Theorem 1 makes the family

$$\{f_{i,j} : 1 \leq i \leq N, 1 \leq j \leq d_i\}$$

a global orthonormal frame on $\mathcal{U}$ under the metric g . Note, however, that this Assumption is not too restrictive and is only made for convenience. Indeed, by restricting the submanifold $\mathcal{U} \subset \mathcal{M}$ if needed, we may always select a set of independent vector fields that span each distribution $\mathcal{E}_i$.

Next, we proceed to define $\{\vartheta^{i,j}\}$ as the co-frame dual to

$$\{f_{i,j} : 1 \leq i \leq N, 1 \leq j \leq d_i\},$$

i.e., the unique set of differential 1-forms on $\mathcal{U}$ that satisfy

$$\vartheta^{i,j}[f_{k,\ell}] = \delta_{ik} \delta_{j\ell}. \quad (17)$$

With this definition, the metric and the orthogonal projection operators onto the distributions $\mathcal{E}_i$ take the form

$$g(v, w) = \sum_{i,j} \vartheta^{i,j}[v] \vartheta^{i,j}[w], \quad P_{\mathcal{E}_i} = \sum_{j=1}^{d_i} \vartheta^{i,j} \otimes f_{i,j}. \quad (18)$$

In addition, for any smooth function J ,

$$\mathbf{grad}(J) = \sum_{i=1}^N \mathbf{grad}_{\mathcal{E}_i}(J), \quad \mathbf{grad}_{\mathcal{E}_i}(J) = \sum_{j=1}^{d_i} dJ[f_{i,j}] f_{i,j}. \quad (19)$$

We then have the following Theorem.

Theorem 2: With respect to the frame $\{f_{i,j}\}$, the linear map $A|_{x^*}$ admits the block matrix representation

$$A|_{x^*} = [A_{ij}|_{x^*}]_{i,j=1}^N, \quad A_{ij}|_{x^*} \in \mathbb{R}^{d_i \times d_j},$$

where the entries of the matrices $A_{ij}|_{x^*}$ are given by

$$\begin{aligned} (A_{ij}|_{x^*})_{p,q} &= d(dJ_i[f_{i,p}])[f_{j,q}]|_{x^*} \\ &= \mathbf{Hess}(J_i)[f_{i,p}, f_{j,q}]|_{x^*} + dJ_i[\nabla_{f_{j,q}} f_{i,p}]|_{x^*}, \end{aligned} \quad (20)$$

for $p \in \{1, \dots, d_i\}$ and $q \in \{1, \dots, d_j\}$, and where $\mathbf{Hess}(J_i)$ denotes the Hessian of J_i with respect to g . $\square$

Remark 5: Formula (20) splits each block of the Jacobian into a Hessian term and a *connection* term $dJ_i[\nabla_{f_{j,q}} f_{i,p}]$. The connection term encodes the non-holonomy of the actuation frame $\{f_{i,j}\}$ and, in general, it does not vanish at x^* , since (12) does not imply $\mathbf{grad}(J_i)|_{x^*} = 0$. Nevertheless, there are two instructive cases with some simplifications. If the inter-block

connection coefficients vanish, i.e., $\nabla_{f_{j,q}} f_{i,p} \equiv 0$ whenever $i \neq j$, then the off-diagonal blocks are pure Hessian blocks,

$$(A_{ij}|_{x^*})_{p,q} = \mathbf{Hess}(J_i)[f_{i,p}, f_{j,q}]|_{x^*}, \quad i \neq j,$$

while the diagonal blocks retain the term $dJ_i[\nabla_{f_{i,q}} f_{i,p}]|_{x^*}$, which reflects the intra-block geometry of the frame. Conversely, if the intra-block connection coefficients vanish, i.e., $\nabla_{f_{i,q}} f_{i,p} \equiv 0$ for all i , then the diagonal blocks are pure Hessian blocks, while the off-diagonal blocks retain the term $dJ_i[\nabla_{f_{j,q}} f_{i,p}]|_{x^*}$ for $i \neq j$, which reflects the inter-block geometry of the frame. Finally, if the frame is *holonomic*, i.e., $[f_{i,p}, f_{j,q}] \equiv 0$ for all pairs, then the frame is a coordinate frame in some chart and the metric constructed in Theorem 1 is flat on that chart. In particular, all connection terms vanish identically, and, in this case, the blocks (20) reduce to matrices of mixed partial derivatives in this chart, i.e.,

$$(A_{ij}|_{x^*})_{p,q} = \partial_{(j,q),(i,p)}^2 J_i|_{x^*},$$

which recovers the characterization in [8]. In general, however, the terms induced by the non-holonomy of the frame $\{f_{i,j}\}$ must be taken into account in studying (3). $\square$

Before we move on from the current section, we provide examples that illustrate the concepts introduced thus far.

Example 4 (Example 1 cont'd): With the vector fields $\{e_i\}$ defined in Example 1, we define the co-vector fields $\{\vartheta^i\}$ as

$$\vartheta^i(x)[v] := (-1)^{3-i} \Delta(x)^{-1} \langle v, e_{3-i}(x) \times x \rangle_{\mathbb{R}^3},$$

for all $(x, v) \in T\mathcal{U}$. Then, it can be verified through direct computation that $\vartheta^i[e_j] = \delta_{ij}$, i.e., $\{\vartheta^i\}$ is the dual co-frame to the frame $\{e_i\}$. We now define a metric g on the submanifold $\mathcal{U}$ by (18) for any pair of vector fields v, w . Next, we introduce the payoff functions that characterize the decision-making problem:

$$J_i(x) = 2^{\frac{1}{2}} + (x_3 + x_{3-i}),$$

which are depicted in Figure 4. Direct computation gives

$$dJ_i[e_i](x) = -x_i. \quad (21)$$

Therefore, the only equilibrium point in $\mathcal{U}$ is given by $x^* = (0, 0, 1)$. To verify the structural, as well as the asymptotic, stability properties of x^* , we need to compute the intrinsic Jacobian A defined in (13). To that end, we invoke Theorem 2 and utilize the simplified expression obtained in (20). Substituting from (21) shows that $d(dJ_i[e_i])[e_j]|_{x^*} = -\delta_{ij}$, i.e., the matrix representation of $A|_{x^*}$ with respect to the frame $\{e_1, e_2\}$ is given by:

$$A|_{x^*} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

which is clearly an automorphism of $T_{x^*}\mathcal{U}$ and also Hurwitz. It follows from Proposition 1 that x^* is a non-degenerate isolated locally exponentially stable equilibrium. $\square$

Example 5 (Example 2 cont'd): Next, we define the co-vector fields ϑ^i by

$$\vartheta^i(x)[v] := \frac{v_i}{x_1 x_2 x_3},$$

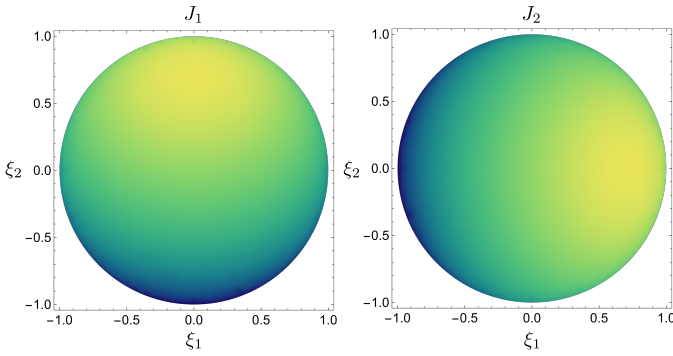


Fig. 4. An illustration of the payoffs J_i for Example 1 in the ξ -coordinates. The red dots represent the maximizer of the corresponding function.

for any tangent vector $v = (v_1, v_2, v_3) \in T_x \mathcal{U}$. Then, it can be verified through direct computation that $\vartheta^i[e_j] = \delta_{ij}$, i.e., the co-frame $\{\vartheta^i\}$ is dual to the frame $\{e_i\}$. We now define a metric g on the submanifold $\mathcal{U}$ by (18), which, automatically, makes the frame $\{e_i\}$ orthonormal with respect to g . The payoff functions of the agents are

$$J_i(x) := 1 - \frac{1}{2}((x_{3-i} - \alpha)^2 + (x_i - 1 + 2\alpha)^2 + (x_3 - \alpha)^2),$$

for any $\alpha \in \mathbb{R}$, which are depicted in Figure 5 for $\alpha = 4/9$. Direct computation gives that

$$dJ_i[e_i](x) = -x_1 x_2 x_3 (x_i - x_3 + 3\alpha - 1), \quad (22)$$

which can be shown to imply that the only equilibrium in $\mathcal{U}$ is the point

$$x^* = \frac{1}{3}(2 - 3\alpha, 2 - 3\alpha, 6\alpha - 1),$$

provided that $\alpha \in (1/6, 2/3)$. To verify the structural, as well as the asymptotic, stability properties of x^* , we compute the intrinsic Jacobian A by invoking Theorem 2 and using (20). Indeed, substituting from (22) into (20), we obtain:

$$\begin{aligned} d(dJ_1[e_1])[e_1] &= x_1 x_2^2 x_3 (x_1 (3\alpha - 4x_3 - 1) \\ &\quad + x_3 (-3\alpha + x_3 + 1) + x_1^2), \\ d(dJ_1[e_1])[e_2] &= x_1^2 x_2 x_3 (x_2 (3\alpha + x_1 - 2x_3 - 1) \\ &\quad + x_3 (-3\alpha - x_1 + x_3 + 1)), \\ d(dJ_1[e_2])[e_1] &= x_1 x_2^2 x_3 (x_1 (3\alpha + x_2 - 2x_3 - 1) \\ &\quad + x_3 (-3\alpha - x_2 + x_3 + 1)), \\ d(dJ_1[e_2])[e_2] &= x_1^2 x_2 x_3 (x_2 (3\alpha - 4x_3 - 1) \\ &\quad + x_3 (-3\alpha + x_3 + 1) + x_2^2), \end{aligned}$$

so that, at the point x^* , we have that the matrix representation of the linear map $A|_{x^*}$ is

$$A|_{x^*} = \frac{1}{729}(1 - 6\alpha)^2(2 - 3\alpha)^4 \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix},$$

which is clearly an automorphism of $T_{x^*} \mathcal{U}$ and Hurwitz. It follows from Proposition 1 that x^* is a non-degenerate isolated locally exponentially stable equilibrium for the system. $\square$

To end this section, we consider additional assumptions on the family of vector fields and investigate the resulting simplifications on the geometric structures introduced earlier. We

begin with the case in which each agent's control distribution is involutive.

Assumption 3: Suppose that, for each $i \in \{1, \dots, N\}$, $[f_{i,p}, f_{i,q}] \in \text{span}\{f_{i,r}\}_{r=1}^{d_i}$, for all $p, q \in \{1, \dots, d_i\}$. $\square$

In words, Assumption 3 asks that each agent cannot directly interfere in the directions influenced by other agents by exciting Lie Brackets of its own vector fields, which is a natural Assumption. Note that Assumption 3 holds trivially in Examples 1 and 2, since each distribution therein is of rank one. Under Assumptions 1 and 3, each distribution $\mathcal{E}_i$ defines a submanifold $\mathcal{U}_i$ such that $T_x \mathcal{U}_i = \mathcal{E}_i(x)$. Note, however, that this is not enough to conclude that $\mathcal{U}$ admits a product structure $\mathcal{U} = \mathcal{U}_1 \times \dots \times \mathcal{U}_N$. Indeed, as in Examples 1 and 2, each distribution $\mathcal{E}_i$ can be 1-dimensional, which is trivially involutive, and yet the manifold $\mathcal{U}$ does not admit a product structure that respects the splitting in Assumption 1 due to the fact that vector fields belonging to different distributions may not commute. Nevertheless, Assumption 3 allows us to simplify the expressions (20) as follows.

First, by Frobenius Theorem [29, Theorem 19.12], there exists an embedded submanifold $\mathcal{U}_i \subset \mathcal{U}$ such that $T\mathcal{U}_i = \mathcal{E}_i$, for all i . That is, $\mathcal{U}_i \subset \mathcal{U}$ is a *weakly embedded* submanifold [29, Theorem 19.17] of the Riemannian manifold $\mathcal{U}$ and, when equipped with the ambient metric, is isometrically weakly embedded. Consequently, restrictions of smooth functions to $\mathcal{U}_i$ are also smooth functions. Introducing the notation $J_i^{\mathcal{U}_i}$ to denote the restriction of the smooth function J_i to the submanifold $\mathcal{U}_i$, as well as g^i , ∇^i , to denote the Riemannian metric on $\mathcal{U}_i$ induced by the ambient metric g , and its associated Levi-Civita connection, respectively, we have the following Proposition.

Proposition 3: Suppose that Assumptions 1-3 hold, and that $x^* \in \mathcal{U}$ satisfies (12). Then, with respect to the frame $\{f_{i,j}\}$, the diagonal blocks in (20) are given by

$$(A_{ii}|_{x^*})_{p,q} = \text{Hess}^i(J_i^{\mathcal{U}_i})[f_{i,p}, f_{i,q}]|_{x^*}, \quad (23)$$

where Hess^i is the intrinsic Hessian of $J_i^{\mathcal{U}_i}$, i.e. the restriction of the function J_i to the submanifold $\mathcal{U}_i$, computed with respect to the induced metric g^i . $\square$

Remark 6: It is worth mentioning that a major simplification provided by Proposition 3 is the fact that the matrix $A_{ii}|_{x^*}$ is now symmetric, since the Hessian is a symmetric operator. We also remark that Assumption 3 will no longer hold, in general, if deception is involved. Indeed, as shown later in Proposition 5, the effect of deception in the presence of information asymmetry is to tilt the distributions $\mathcal{E}_i$ controlled by the oblivious agents along the distributions controlled by the deceiver. Therefore, as soon as deception is involved, the new distributions $\tilde{\mathcal{E}}_i$ may no longer be involutive in the general case. This is a major motivation behind considering the general analysis presented in this section, which allows us to study the stability of deception in Section V. $\square$

Next, we consider the following Assumption.

Assumption 4: For all $i, j \in \{1, \dots, N\}$ with $i \neq j$,

$$[f_{i,p}, f_{j,q}] = 0,$$

for all $p \in \{1, \dots, d_i\}$ and all $q \in \{1, \dots, d_j\}$.

In words, Assumption 4 requires that the vector fields controlled by each agent commute with those controlled by every other agent. Note that Assumption 4 fails in both Examples 1 and 2, in view of the non-vanishing commutators computed therein, which illustrates that the product-structure setting considered above is genuinely more restrictive. Under Assumptions 1-4, Frobenius Theorem can be applied to show the existence of a local diffeomorphism $\phi : U \rightarrow \mathcal{U}_1 \times \dots \times \mathcal{U}_N$, where each $\mathcal{U}_i$ is a smooth d_i -dimensional Riemannian manifold and $\phi_* \mathcal{E}_i = T\mathcal{U}_i$. In particular, the Riemannian metric g from Theorem 1 takes now a block-diagonal form $g = g^1 \oplus \dots \oplus g^N$ in these coordinates.

The following proposition establishes the key geometric consequence of the combined assumptions: the Levi-Civita connection of g has no inter-agent components.

Proposition 4: Let Assumptions 1-4 be satisfied. Then,

$$\nabla_{f_{j,q}} f_{i,p} = 0 \quad \text{on } U, \quad \text{for all } i \neq j. \quad (24)$$

□

As a consequence of Proposition 4, the intrinsic Jacobian from Theorem 2 simplifies completely.

Corollary 1: Let the hypotheses of Proposition 4 hold and let $x^* \in \mathcal{U}$ satisfy (12). Then, the blocks of the intrinsic Jacobian $A|_{x^*}$ are given by

$$(A_{ij}|_{x^*})_{p,q} = \begin{cases} \text{Hess}^i(J_i^{\mathcal{U}_i})[f_{i,p}, f_{i,q}]|_{x^*}, & i = j, \\ \text{Hess}(J_i)[f_{i,p}, f_{j,q}]|_{x^*}, & i \neq j. \end{cases} \quad (25)$$

□

Proof: For diagonal blocks, the expression in (25) follows from Proposition 3, which requires only Assumptions 1-3. For off-diagonal blocks with $i \neq j$, substituting (24) into the second equality of (20) immediately gives $(A_{ij}|_{x^*})_{p,q} = \text{Hess}(J_i)[f_{i,p}, f_{j,q}]|_{x^*}$. ■

Remark 7: In the special case where each $\mathcal{E}_i$ is spanned by coordinate vector fields in some chart, i.e., $f_{i,j} = \partial_{(i,j)}$, Assumptions 2-4 are automatically satisfied, and system (3) takes the form

$$\dot{x} = \sum_{i=1}^N \sum_{j=1}^{d_i} \partial_{(i,j)} J_i(x) \partial_{(i,j)},$$

in coordinates, i.e., the interaction is governed by the vector field whose components are partial derivatives of the payoff functions in local coordinates. Moreover, we compute that

$$\begin{aligned} \text{Hess}^i(J_i)[f_{i,p}, f_{i,q}] &= \partial_{(i,p),(i,q)}^2 J_i, \\ \text{Hess}(J_i)[f_{i,p}, f_{j,q}] &= \partial_{(j,q),(i,p)}^2 J_i. \end{aligned}$$

In this case, (25) recovers the coordinate dependent formulas studied in [8], [9] as a special case of Theorem 2. Note, however, that even if Assumptions 2-4 hold, the results in [8], [9] still do not apply due to the fact that, for each i , the commutators $[f_{i,p}, f_{i,q}]$ need not vanish in general. In particular, the expression (25) accounts for the intrinsic curvature of the submanifolds $\mathcal{U}_i$ which, in general, does not vanish. □

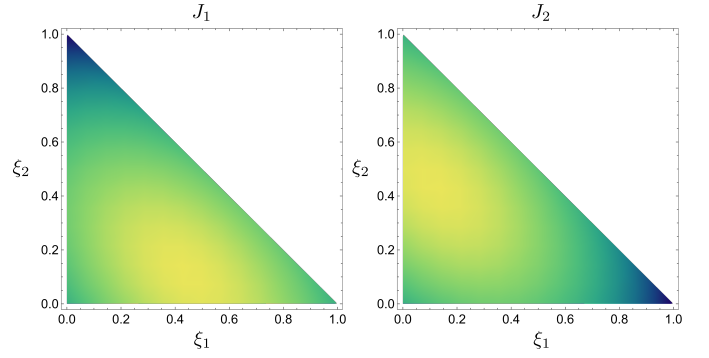


Fig. 5. An illustration of payoffs J_i for Example 2 in the ξ -coordinates. The red dots represent the maximizer of the corresponding function.

V. INFORMATION ASYMMETRY AND DECEPTION

As discussed in Section III, when the assumption of information symmetry is violated, an agent with access to privileged information can bias the steady-state behavior of (1) by manipulating only its own control inputs, without interfering with the actions or measurements of the oblivious agents. In this section, we analyze the interaction dynamics induced by the deceptive feedback law (6), i.e., system (8), repeated below for convenience,

$$\dot{x} = \sum_{i=1}^2 \kappa_i \sum_{j=1}^{d_i} L_{\tilde{f}_{i,j}}(J_i)(x) \tilde{f}_{i,j}(x), \quad (8)$$

where, as in (7b), the vector fields $\tilde{f}_{i,j}$ are given by

$$\tilde{f}_{1,j} = f_{1,j}, \quad \tilde{f}_{2,j} := f_{2,j} + \sum_{k=1}^{d_1} \gamma_j^k f_{1,k}.$$

Similar to Section IV, we impose Assumption 1 on the vector fields $f_{i,j}$ and, for simplicity, we also impose Assumption 2. Then, we have the following proposition.

Proposition 5: Let $\{\tilde{f}_{i,j}\}$ be given by (7b) for any choice of deception parameters $\{\gamma_j^k\}$. Then, Assumption 1 is satisfied on $\mathcal{U}$ with $\tilde{\mathcal{E}}_i = \text{span}\{\tilde{f}_{i,j}\}_{j=1}^{d_i}$. In particular, the family $\{\tilde{f}_{i,j}\}$ remains pointwise linearly independent on $\mathcal{U}$ and

$$T_x \mathcal{U} = \tilde{\mathcal{E}}_1(x) \oplus \tilde{\mathcal{E}}_2(x)$$

for all $x \in \mathcal{U}$. □

Remark 8 (The geometric meaning of deception): Although $\tilde{\mathcal{E}}_1 = \mathcal{E}_1$ as subbundles of $T\mathcal{U}$, since the deceiver does not alter their own directions, the victim's new bundle $\tilde{\mathcal{E}}_2$ is generically distinct from $\mathcal{E}_2$ since, from (7b), the vector field $f_{2,j}$ acquires components along the deceiver's directions $f_{1,k}$ with weights γ_j^k . This is the geometric mechanism of deception in our setting: the deceiver *rotates the victim's effective actuation directions* into its own bundle without modifying its own reach. As a consequence, projected gradients on $\tilde{\mathcal{E}}_2$ are biased relative to those on $\mathcal{E}_2$, and the equilibrium of the averaged dynamics is generically shifted. Moreover, the proposition holds for *all* choices of deception parameters, including arbitrarily large ones, and uniformly in $x \in \mathcal{U}$. This is a direct consequence of the block-triangular

structure of Γ in (50), which decouples the invertibility analysis from the magnitude of the tilt block γ . $\square$

As a direct consequence of Theorem 1 and Proposition 5, we have the following Corollary.

Corollary 2: There exists a Riemannian metric $\tilde{g}$ on the open sub-manifold $\mathcal{U} \subset \mathcal{M}$ such that (8) takes the form

$$\dot{x} = \sum_{i=1}^2 \kappa_i \mathbf{grad}_{\tilde{\mathcal{E}}_i}(J_i)(x), \quad (26)$$

where $\mathbf{grad}_{\tilde{\mathcal{E}}_i}(J_i)$ is the vector field obtained through orthogonal projection of $\mathbf{grad}(J_i)$ with respect to $\tilde{g}$ onto $\tilde{\mathcal{E}}_i$. $\square$

The result of Corollary 2 allows us to, once again, take $\kappa_i = 1$ and look for equilibria of system (8) using the condition

$$\mathbf{grad}_{\tilde{\mathcal{E}}_i}(J_i)(x^*) = 0, \quad i \in \{1, 2\}. \quad (27)$$

In particular, since $\tilde{\mathcal{E}}_1 = \mathcal{E}_1$, this condition is unchanged from the analysis in Section IV for $i = 1$. However, unlike Section IV, the bundle $\tilde{\mathcal{E}}_2$ is tilted along the bundle $\mathcal{E}_1$. In particular, utilizing the assumption that the vector fields $\{f_{i,j}\}$, and, consequently the vector fields $\{\tilde{f}_{i,j}\}$, are linearly independent, as well as (7b), the new equilibrium conditions take the form

$$\begin{aligned} 0 &= dJ_1[f_{1,j}](x), & j &\in \{1, \dots, d_1\}, \\ 0 &= dJ_2[f_{2,j}](x) + \sum_{k=1}^{d_1} \gamma_j^k dJ_2[f_{1,k}](x), & j &\in \{1, \dots, d_2\}, \end{aligned}$$

which implies that an equilibrium point of system (8) will, in general, depend on the matrix of deception parameters $\gamma \in \mathbb{R}^{d_1 \times d_2}$. Nevertheless, we have the following Theorem showing that the stability properties of this equilibrium point are preserved for small values of γ :

Theorem 3: Suppose that $x^* \in \mathcal{U}$ is a locally exponentially stable equilibrium point for the interaction dynamics (3). Then, there exists an open neighborhood $\Omega \subset \mathbb{R}^{d_1 \times d_2}$ with $0 \in \Omega$ and a smooth map $X^* : \Omega \rightarrow \mathcal{U}$ such that $X^*(0) = x^*$ and, for all $\gamma \in \Omega$, the point $X^*(\gamma)$ is a locally exponentially stable equilibrium point for the interaction dynamics (8). $\square$

Corollary 3: Under the hypotheses of Theorem 3, and under the identification of $T_{x^*}\mathcal{U}$ with $\mathbb{R}^n$ provided by the frame $\{f_{i,j}(x^*)\}$, the map X^* satisfies

$$\left. \frac{\partial X^*}{\partial \gamma_j^k} \right|_{\gamma=0} = -dJ_2[f_{1,k}](x^*) (A|_{x^*})^{-1} [f_{2,j}(x^*)], \quad (28)$$

for all $j \in \{1, \dots, d_2\}$ and $k \in \{1, \dots, d_1\}$. $\square$

Remark 9: Formula (28) makes the mechanism of Proposition 5 quantitative: to first order, the deceiver shifts the equilibrium along the directions $(A|_{x^*})^{-1}[f_{2,j}(x^*)]$, weighted by the sensitivities $dJ_2[f_{1,k}](x^*)$ of the victim's payoff along the deceiver's directions. In particular, if $dJ_2[f_{1,k}](x^*) = 0$ for all k , i.e., if the victim's payoff is stationary along the deceiver's directions at x^* , then deception is ineffective to first order, regardless of the choice of γ . Formula (28) thus provides a natural starting point for the constructive design of deception parameters in multi-agent adaptive systems with

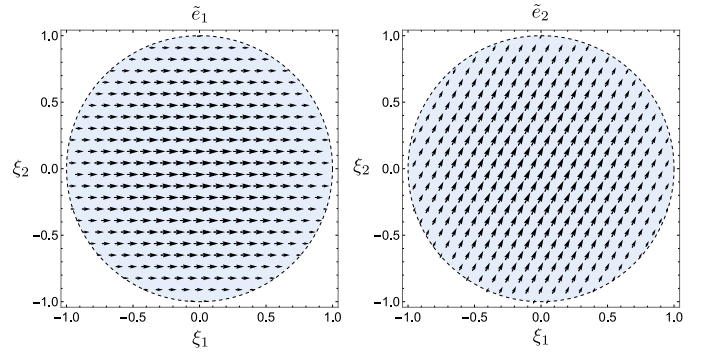


Fig. 6. An illustration of the tilted vector fields $\tilde{e}_i$ for Example 6 in the ξ -coordinates. The arrows' sizes represent the magnitude of the vector field.

asymmetric information, a direction to be pursued in future work. $\square$

Remark 10: The restriction to two agents considered in this section is made for clarity of exposition, not out of necessity. In particular, consider N agents and suppose that the information-asymmetry structure is *acyclic*, i.e., there exists an ordering of the agents such that agent i can measure, and embed into its own inputs, the inputs of agent ℓ only if $i < \ell$. Then, the matrix Γ representing the tilt, cf. (50), is strictly block upper-triangular in this ordering, hence nilpotent, and $\det(I + \Gamma) = 1$ irrespective of the deception parameters. Consequently, Proposition 5, Theorem 3, and Corollary 3 extend verbatim to this setting. The genuinely distinct case is that of a *cyclic* information structure, e.g., two agents simultaneously attempting to deceive one another, in which case $I + \Gamma$ may become singular for certain parameter choices and the tilted family may fail to span the tangent space; the characterization of equilibria in this regime is an open problem that we leave for future work. $\square$

We conclude this section by revisiting the examples from Section IV when deception is involved.

Example 6 (Example 1 cont'd): We consider the case in which the 1st-agent implements a deceptive strategy towards the 2nd-agent. In this case, $\{e_i\}$ are replaced by

$$\tilde{e}_1(x) := e_1(x), \quad \tilde{e}_2(x) = e_2(x) + \gamma e_1(x),$$

where $\gamma \in \mathbb{R}$ is the deception parameter. The new vector fields are depicted in Figure 6 for $\gamma = \frac{1}{2}$. As can be observed from the figure, the vector field $\tilde{e}_2$ is tilted compared to Figure 2. Repeating the same computations as in Example 1, we find that the new frame $\{\tilde{e}_i\}$, with $\tilde{e}_1 = e_1$ and $\tilde{e}_2$ as above, still spans the tangent space of the submanifold $\mathcal{U}$. In particular, we have that

$$dJ_1[e_1](x) = -x_1, \quad dJ_2[\tilde{e}_2] = -x_2 - \gamma(x_1 - x_3),$$

so that the only equilibrium in $\mathcal{U}$ is given, as a function of the deception parameter γ , by

$$X^*(\gamma) = \left(0, \frac{\gamma}{\sqrt{1+\gamma^2}}, \frac{1}{\sqrt{1+\gamma^2}}\right). \quad (29)$$

Next, we proceed to verify the structural, as well as the asymptotic, stability properties of the point $X^*(\gamma)$. Thanks

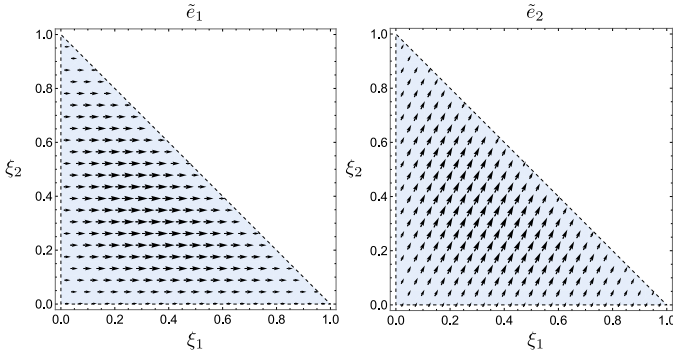


Fig. 7. An illustration of the tilted vector fields $\tilde{e}_i$ for Example 7 in the ξ -coordinates. The arrows' sizes represent the magnitude of the vector field.

to Theorem 3, we already know that there exists an open neighborhood $\Omega \subset \mathbb{R}$ with $0 \in \Omega$ such that, for all $\gamma \in \Omega$, $X^*(\gamma)$ is locally exponentially stable. Nevertheless, because we have an explicit expression of the equilibrium point, we can obtain stronger results. To do so, we employ the frame $\{\tilde{e}_1, \tilde{e}_2\}$, which is orthonormal with respect to the tilted metric $\tilde{g}$ of Corollary 2. In particular, we compute that

$$\begin{aligned} d(dJ_1[\tilde{e}_1])[\tilde{e}_1] &= -x_3, \\ d(dJ_1[\tilde{e}_1])[\tilde{e}_2] &= -\gamma x_3, \\ d(dJ_2[\tilde{e}_2])[\tilde{e}_1] &= -\gamma(x_1 + x_3), \\ d(dJ_2[\tilde{e}_2])[\tilde{e}_2] &= -\gamma(\gamma x_1 + x_2) - (1 + \gamma^2)x_3. \end{aligned}$$

Hence, substituting from (29), we see that the matrix representation of $A|_{X^*(\gamma)}$ with respect to $\{\tilde{e}_i\}$ is

$$A|_{X^*(\gamma)} = \frac{1}{\sqrt{1+\gamma^2}} \begin{bmatrix} -1 & -\gamma \\ -\gamma & -(1+2\gamma^2) \end{bmatrix}.$$

To verify that $A|_{X^*(\gamma)}$ is an automorphism of $T_{X^*(\gamma)}\mathcal{U}$ and Hurwitz, we compute its trace and determinant as:

$$\begin{aligned} \text{trace}(A|_{X^*(\gamma)}) &= -2\sqrt{1+\gamma^2} \\ \det(A|_{X^*(\gamma)}) &= 1, \end{aligned}$$

which, since it is a 2×2 matrix, imply that its eigenvalues are strictly negative, and, therefore, $X^*(\gamma)$ is a non-degenerate isolated exponentially stable equilibrium for all $\gamma \in \mathbb{R}$. In conclusion, deception preserves the stability of the new equilibrium point (29), which is controlled by the deceiver via the parameter γ . We also note that differentiating (29) at $\gamma = 0$ gives $\frac{d}{d\gamma}X^*(0) = (0, 1, 0) = e_2(x^*)$, in agreement with Corollary 3, since $dJ_2[e_1](x^*) = 1$ and $A|_{x^*} = -I$. $\square$

Example 7 (Example 2 cont'd): We consider the case in which the 1st-agent implements a deceptive strategy towards the 2nd-agent in the setting of Example 2. In this case, the vector fields $\{e_i\}$ are replaced by

$$\tilde{e}_1(x) := e_1(x), \quad \tilde{e}_2(x) := e_2(x) + \gamma e_1(x),$$

where $\gamma \in \mathbb{R}$ is the deception parameter. The new vector fields are depicted in Figure 7 for $\gamma = \frac{1}{2}$. As can be observed from the figure, the vector field $\tilde{e}_2$ is tilted compared to Figure 3. We compute that

$$dJ_2[e_1](x) = x_1 x_2 x_3 (x_3 - x_1),$$

and, consequently,

$$dJ_1[\tilde{e}_1](x) = x_1 x_2 x_3 (-x_1 + x_3 + 1 - 3\alpha),$$

$$dJ_2[\tilde{e}_2](x) = x_1 x_2 x_3 [(-x_2 + x_3 + 1 - 3\alpha) + \gamma(x_3 - x_1)],$$

where we utilized (22). On $\mathcal{U}$ we have $x_1 x_2 x_3 > 0$, so the equilibrium conditions reduce to the two linear equations

$$\begin{aligned} -x_1 + x_3 + 1 - 3\alpha &= 0, \\ (-x_2 + x_3 + 1 - 3\alpha) + \gamma(x_3 - x_1) &= 0, \end{aligned}$$

together with the simplex constraint $x_1 + x_2 + x_3 = 1$. Introducing $\beta := 1 - 3\alpha$, the first equation gives $x_1 = x_3 + \beta$, so that $x_3 - x_1 = -\beta$. Substituting into the second and simplifying yields $x_2 = x_3 + \beta(1 - \gamma)$. Therefore, the simplex constraint determines

$$x_3 = \frac{(6\alpha - 1) + \gamma(1 - 3\alpha)}{3}.$$

From the above, we see that the unique equilibrium in $\mathcal{U}$, when it exists as a function of the deception parameter γ , is

$$X^*(\gamma) = \frac{1}{3} \begin{pmatrix} (2 - 3\alpha) + \gamma(1 - 3\alpha) \\ (2 - 3\alpha) - 2\gamma(1 - 3\alpha) \\ (6\alpha - 1) + \gamma(1 - 3\alpha) \end{pmatrix}. \quad (30)$$

In particular, $X^*(0)$ recovers the nominal equilibrium from Example 2, and the γ -dependent shift

$$X^*(\gamma) - X^*(0) = \frac{\gamma(1 - 3\alpha)}{3} (1, -2, 1)$$

is exactly linear in γ . For $X^*(\gamma)$ to remain in the interior $\mathcal{U}$, we require $\gamma \in \Omega$, where $\Omega \subset \mathbb{R}$ is the open interval

$$\Omega = \{\gamma \in \mathbb{R} : X_i^*(\gamma) > 0 \text{ for } i = 1, 2, 3\}. \quad (31)$$

The three positivity constraints in (31) read $(2 - 3\alpha) + \gamma(1 - 3\alpha) > 0$, $(2 - 3\alpha) - 2\gamma(1 - 3\alpha) > 0$, $(6\alpha - 1) + \gamma(1 - 3\alpha) > 0$. For $\alpha \in (1/6, 1/3)$, these give that

$$\Omega = \left(\frac{1 - 6\alpha}{1 - 3\alpha}, \frac{2 - 3\alpha}{2(1 - 3\alpha)} \right),$$

which contains 0 and is bounded on both sides. For $\alpha = 1/3$, the map $\gamma \mapsto X^*(\gamma)$ is constant ($\beta = 0$), and deception has no effect on the equilibrium. For $\alpha \in (1/3, 2/3)$ (so $1 - 3\alpha < 0$), the inequalities flip direction and

$$\Omega = \left(\frac{2 - 3\alpha}{2(1 - 3\alpha)}, \frac{1 - 6\alpha}{1 - 3\alpha} \right).$$

In both non-degenerate regimes, Ω is an explicit, bounded, open interval containing 0, giving a quantitative counterpart to the qualitative neighborhood guaranteed by Theorem 3.

To verify local exponential stability of $X^*(\gamma)$, we compute the intrinsic Jacobian $A|_{X^*(\gamma)}$ with respect to the frame $\{\tilde{e}_1, \tilde{e}_2\}$, which is orthonormal with respect to the tilted metric $\tilde{g}$, via Theorem 2.

Observe that each directional derivative above factors as $\chi \cdot \rho$, where $\chi := x_1 x_2 x_3$ and ρ is a polynomial of degree 1 in x . At $X^*(\gamma)$, the linear factor ρ vanishes by construction of the

equilibrium, while $\chi|_{X^*(\gamma)} > 0$. Hence, using $\tilde{e}_1 = \chi(\partial_1 - \partial_3)$ and $\tilde{e}_2 = \chi(\gamma\partial_1 + \partial_2 - (1 + \gamma)\partial_3)$,

$$d(dJ_i[\tilde{e}_i])[\tilde{e}_j]|_{X^*(\gamma)} = \chi \cdot \tilde{e}_j[\rho_i]|_{X^*(\gamma)},$$

where $\rho_1 = -x_1 + x_3 + \beta$ and $\rho_2 = (-x_2 + x_3 + \beta) + \gamma(x_3 - x_1)$. Direct computation gives

$$\begin{aligned} d(dJ_1[\tilde{e}_1])[\tilde{e}_1] &= -2\chi^2, \\ d(dJ_1[\tilde{e}_1])[\tilde{e}_2] &= -(1 + 2\gamma)\chi^2, \\ d(dJ_2[\tilde{e}_2])[\tilde{e}_1] &= -(1 + 2\gamma)\chi^2, \\ d(dJ_2[\tilde{e}_2])[\tilde{e}_2] &= -2(1 + \gamma + \gamma^2)\chi^2. \end{aligned}$$

Hence, substituting from (30), the matrix representation of $A|_{X^*(\gamma)}$ with respect to $\{\tilde{e}_i\}$ is

$$A|_{X^*(\gamma)} = \chi^2|_{X^*(\gamma)} \begin{bmatrix} -2 & -(1 + 2\gamma) \\ -(1 + 2\gamma) & -2(1 + \gamma + \gamma^2) \end{bmatrix}. \quad (32)$$

In particular, at $\gamma = 0$, the scalar $\chi^2|_{x^*} = \frac{1}{729}(2 - 3\alpha)^4(6\alpha - 1)^2$ and the matrix (32) reduces to the expression in Example 5. Next, we verify that $A|_{X^*(\gamma)}$ is Hurwitz by computing its trace and determinant:

$$\begin{aligned} \text{tr}(A|_{X^*(\gamma)}) &= -2\chi^2|_{X^*(\gamma)} \cdot (2 + \gamma + \gamma^2), \\ \det(A|_{X^*(\gamma)}) &= \chi^4|_{X^*(\gamma)} \cdot [4(1 + \gamma + \gamma^2) - (1 + 2\gamma)^2] \\ &= 3\chi^4|_{X^*(\gamma)}. \end{aligned}$$

Since $2 + \gamma + \gamma^2 = (\gamma + 1/2)^2 + 7/4 > 0$ for all $\gamma \in \mathbb{R}$, the trace is strictly negative for every $\gamma \in \Omega$. The determinant equals $3\chi^4|_{X^*(\gamma)} > 0$ for every $\gamma \in \Omega$. A real 2×2 matrix with strictly negative trace and strictly positive determinant is Hurwitz, and therefore $X^*(\gamma)$ is a non-degenerate isolated locally exponentially stable equilibrium for system (8), for every $\gamma \in \Omega$. In conclusion, as in Example 6, deception preserves the stability of the system on the entire admissible parameter interval Ω , with the equilibrium smoothly controlled by γ according to (30) with $\gamma \in \Omega$. $\square$

VI. PROOFS

A. Proof of Theorem 1

Under Assumption 1, and for each $x \in \mathcal{U}$, we have the (pointwise) direct-sum decomposition $T_x\mathcal{U} = \mathcal{E}_1(x) \oplus \cdots \oplus \mathcal{E}_N(x)$, where, thanks to Assumption 1, each $\mathcal{E}_i$ is a regular distribution on $\mathcal{U}$ generated by the family $\{f_{i,j}\}_{j=1}^{d_i}$. Consequently, using annihilators yields the canonical dual decomposition

$$T_x^*\mathcal{U} = \mathcal{E}_1(x)^\circ \oplus \cdots \oplus \mathcal{E}_N(x)^\circ, \quad (33)$$

$$\mathcal{E}_i(x)^\circ := \text{Ann}\left(\bigoplus_{k \neq i} \mathcal{E}_k(x)\right) \subset T_x^*\mathcal{U}. \quad (34)$$

Let $\pi_i(x) : T_x^*\mathcal{U} \rightarrow \mathcal{E}_i(x)^\circ$ denote the corresponding linear projections. For each $i \in \{1, \dots, N\}$, define a linear operator

$$P_i(x)[\omega] := \sum_{j=1}^{d_i} (\pi_i(x)\omega)(f_{i,j}(x)) f_{i,j}(x), \quad (35)$$

for all $\omega \in T_x^*\mathcal{U}$. By construction, $P_i(x)[\omega] \in \mathcal{E}_i(x)$ for all $\omega \in T_x^*\mathcal{U}$. Moreover, since $dJ_i(x)(f_{i,j}(x)) = L_{f_{i,j}}(J_i)(x)$, system (3) can be rewritten as

$$\dot{x} = \sum_{i=1}^N \kappa_i P_i(x)[dJ_i(x)]. \quad (36)$$

We now construct a metric g on $\mathcal{U}$ via its co-metric. For each $x \in \mathcal{U}$, we define a symmetric bilinear form g_x^{-1} on $T_x^*\mathcal{U}$ by declaring it block-diagonal with respect to the splitting (33) and, for $\alpha, \beta \in T_x^*\mathcal{U}$ with unique decompositions $\alpha = \sum_{i=1}^N \alpha_i$ and $\beta = \sum_{i=1}^N \beta_i$ such that $\alpha_i, \beta_i \in \mathcal{E}_i(x)^\circ$, defining

$$\begin{aligned} g_x^{-1}(\alpha, \beta) &:= \sum_{i=1}^N g_{i,x}^{-1}(\alpha_i, \beta_i), \\ g_{i,x}^{-1}(\alpha_i, \beta_i) &:= \sum_{j=1}^{d_i} \alpha_i(f_{i,j}(x)) \beta_i(f_{i,j}(x)). \end{aligned} \quad (37)$$

Since $\{f_{i,j}(x)\}_{j=1}^{d_i}$ spans $\mathcal{E}_i(x)$, the quadratic form $\alpha_i \mapsto g_{i,x}^{-1}(\alpha_i, \alpha_i) = \sum_{j=1}^{d_i} \alpha_i(f_{i,j}(x))^2$ is positive definite on $\mathcal{E}_i(x)^\circ$. Hence, g_x^{-1} is positive definite on $T_x^*\mathcal{U}$ for every $x \in \mathcal{U}$. Smoothness follows from the smoothness of the vector fields $f_{i,j}$ and of the projections $\pi_i(x)$ for regular distributions. Therefore, the tensor g^{-1} defines a Riemannian co-metric on $\mathcal{U}$, and, consequently, a unique Riemannian metric g .

Next, let $g_x^\sharp : T_x^*\mathcal{U} \rightarrow T_x\mathcal{U}$ denote the tangent-cotangent isomorphism induced by g , i.e.,

$$\alpha(v) = g_x(g_x^\sharp \alpha, v) \quad \forall \alpha \in T_x^*\mathcal{U}, v \in T_x\mathcal{U}.$$

By construction of (37), we have the block formula

$$g_x^\sharp \alpha = \sum_{i=1}^N P_i(x)[\alpha_i], \quad (38)$$

where $\alpha = \sum_{i=1}^N \alpha_i$, $\alpha_i \in \mathcal{E}_i(x)^\circ$ is the unique decomposition of α with respect to the splitting (33). For each i , the Riemannian gradient of the function J_i with respect to the metric g is defined by $\text{grad}(J_i)(x) := g_x^\sharp(dJ_i(x))$. Utilizing the splitting (33) once more, we have the unique decomposition $dJ_i(x) = \sum_{k=1}^N (dJ_i(x))_k$ with $(dJ_i(x))_k \in \mathcal{E}_k(x)^\circ$, which, in combination with (38), implies that

$$\text{grad}(J_i)(x) = \sum_{k=1}^N P_k(x)[(dJ_i(x))_k].$$

Moreover, since g_x^{-1} is block-diagonal with respect to the splitting (33), it follows that the decomposition (10) is g_x -orthogonal, and, therefore, the g_x -orthogonal projection $P_{\mathcal{E}_i}(x) : T_x\mathcal{U} \rightarrow \mathcal{E}_i(x)$ is precisely the extraction of the i -th component. Hence

$$\text{grad}_{\mathcal{E}_i}(J_i)(x) := P_{\mathcal{E}_i}[\text{grad}(J_i)](x) = P_i(x)[(dJ_i(x))_i].$$

Finally, since $f_{i,j}(x) \in \mathcal{E}_i(x)$ we have that

$$(dJ_i(x))_i(f_{i,j}(x)) = dJ_i(x)(f_{i,j}(x)),$$

and, therefore, it follows that

$$\text{grad}_{\mathcal{E}_i}(J_i)(x) = \sum_{j=1}^{d_i} L_{f_{i,j}}(J_i)(x) f_{i,j}(x).$$

Combining this identity with (36) proves that system (3) is of the form (11), which concludes the proof.

B. Proof of Proposition 2

Let $X := \text{diag}(\chi_1 I_1, \dots, \chi_N I_N)$, where I_i is the identity on $\mathcal{E}_i(x^*)$, and define the operator $\tilde{A} := X^{-1}AX$, which is similar to A , i.e., has the same spectrum. Direct computation shows that its blocks with respect to the splitting (10) are $\tilde{A}_{ii} = A_{ii}$, and $\tilde{A}_{ij} = \frac{\chi_j}{\chi_i} A_{ij}$. Therefore,

$$\tilde{R}_i := \sum_{j \neq i} \|\tilde{A}_{ij}\|_g = \sum_{j \neq i} \|A_{ij}\|_g \frac{\chi_j}{\chi_i},$$

with respect to $\|\cdot\|_g$. By the block Gershgorin circle theorem [34], each $\lambda \in \sigma(A)$ satisfies, for some i ,

$$\|(\tilde{A}_{ii} - \lambda I_i)^{-1}\|_g^{-1} \leq \tilde{R}_i. \quad (39)$$

Since $\tilde{A}_{ii} = A_{ii}$ and $\|M^{-1}\|_g^{-1} = \sigma_{\min}^g(M)$, we obtain that $\sigma_{\min}^g(A_{ii} - \lambda I_i) \leq \tilde{R}_i$. To arrive at a contradiction, suppose that, for some $\lambda \in \sigma(\tilde{A})$, we have that $\Re(\lambda) \geq 0$. Then, we would have that

$$\sigma_{\min}^g(A_{ii} - \lambda I_i) \geq \inf_{\Re(s) \geq 0} \sigma_{\min}^g(A_{ii} - s I_i) > \tilde{R}_i$$

from (16), which contradicts (39). Hence, $\sigma(\tilde{A}) \subset \{\Re(z) < 0\}$ and, because $\sigma(A) = \sigma(\tilde{A})$, A is Hurwitz.

C. Proof of Theorem 2

Recall from (15) that

$$A_{ij}[\cdot] = P_{\mathcal{E}_i} \left[\nabla_{P_{\mathcal{E}_j}[\cdot]} \mathbf{grad}(J_i) \right] + C_{ij}[\cdot], \quad (40)$$

$$C_{ij}[\cdot] = \sum_{k=1}^N P_{\mathcal{E}_i} \left[(\nabla_{P_{\mathcal{E}_j}[\cdot]} P_{\mathcal{E}_k}) [\mathbf{grad}(J_k)] \right]. \quad (41)$$

We evaluate $A_{ij}[v]$ for an arbitrary vector field v and then extract its components in the orthonormal frame $\{f_{i,j}\}$.

From (18) and the product rule, we compute that

$$\nabla P_{\mathcal{E}_k} = \sum_{\ell=1}^{d_k} \left(\nabla \vartheta^{k,\ell} \otimes f_{k,\ell} + \vartheta^{k,\ell} \otimes \nabla f_{k,\ell} \right).$$

Since $P_{\mathcal{E}_j}[v] = \sum_{q=1}^{d_j} \vartheta^{j,q}[v] f_{j,q}$, we obtain that

$$\begin{aligned} \nabla_{P_{\mathcal{E}_j}[v]} P_{\mathcal{E}_k} &= \sum_{q=1}^{d_j} \vartheta^{j,q}[v] \sum_{\ell=1}^{d_k} \left(\vartheta^{k,\ell} \otimes \nabla_{f_{j,q}} f_{k,\ell} \right. \\ &\quad \left. + (\nabla_{f_{j,q}} \vartheta^{k,\ell}) \otimes f_{k,\ell} \right). \end{aligned} \quad (42)$$

From the definition of the covariant derivative of a 1-form,

$$(\nabla_{f_{j,q}} \vartheta^{k,\ell})[\cdot] = d(\vartheta^{k,\ell}[\cdot])[f_{j,q}] - \vartheta^{k,\ell}[\nabla_{f_{j,q}}[\cdot]].$$

Applying this to $\mathbf{grad}(J_k)$ and utilizing (17) and (19),

$$\begin{aligned} (\nabla_{f_{j,q}} \vartheta^{k,\ell})[\mathbf{grad}(J_k)] &= d(dJ_k[f_{k,\ell}])[f_{j,q}] \\ &\quad - \vartheta^{k,\ell}[\nabla_{f_{j,q}} \mathbf{grad}(J_k)]. \end{aligned}$$

Utilizing (19), the product rule, and (17):

$$\begin{aligned} \vartheta^{k,\ell}[\nabla_{f_{j,q}} \mathbf{grad}(J_k)] &= d(dJ_k[f_{k,\ell}])[f_{j,q}] \\ &\quad + \sum_{m,r} dJ_k[f_{m,r}] \vartheta^{k,\ell}[\nabla_{f_{j,q}} f_{m,r}]. \end{aligned}$$

Hence, we obtain that

$$(\nabla_{f_{j,q}} \vartheta^{k,\ell})[\mathbf{grad}(J_k)] = - \sum_{m,r} dJ_k[f_{m,r}] \vartheta^{k,\ell}[\nabla_{f_{j,q}} f_{m,r}]. \quad (43)$$

Substituting (42) into (41) and applying $P_{\mathcal{E}_i}$, which extracts components along $\{f_{i,p}\}_{p=1}^{d_i}$ via (18), we obtain:

$$\begin{aligned} P_{\mathcal{E}_i} \left[(\nabla_{P_{\mathcal{E}_j}[v]} P_{\mathcal{E}_k}) [\mathbf{grad}(J_k)] \right] \\ = \sum_{q,p} \left(\sum_{\ell} dJ_k[f_{k,\ell}] \vartheta^{i,p}[\nabla_{f_{j,q}} f_{k,\ell}] \right) \vartheta^{j,q}[v] f_{i,p} \\ + \sum_{q,\ell} (\nabla_{f_{j,q}} \vartheta^{k,\ell})[\mathbf{grad}(J_k)] \vartheta^{j,q}[v] P_{\mathcal{E}_i}[f_{k,\ell}]. \end{aligned}$$

Using $P_{\mathcal{E}_i}[f_{k,\ell}] = \delta_{ik} f_{i,\ell}$ and substituting (43), the second sum contributes only when $k = i$:

$$- \delta_{ik} \sum_{q,\ell} \left(\sum_{m,r} dJ_k[f_{m,r}] \vartheta^{k,\ell}[\nabla_{f_{j,q}} f_{m,r}] \right) \vartheta^{j,q}[v] f_{i,\ell}.$$

Summing over k and relabeling $\ell \mapsto p$, we obtain that

$$\begin{aligned} C_{ij}[v] &= \sum_{q,p} \left(\sum_{k,\ell} (dJ_k[f_{k,\ell}] \right. \\ &\quad \left. - dJ_i[f_{k,\ell}]) \vartheta^{i,p}[\nabla_{f_{j,q}} f_{k,\ell}] \right) \vartheta^{j,q}[v] f_{i,p}. \end{aligned} \quad (44)$$

From (19) and the product rule,

$$\begin{aligned} \nabla_{f_{j,q}} \mathbf{grad}(J_i) &= \sum_{k,\ell} \left(d(dJ_i[f_{k,\ell}])[f_{j,q}] f_{k,\ell} \right. \\ &\quad \left. + dJ_i[f_{k,\ell}] \nabla_{f_{j,q}} f_{k,\ell} \right). \end{aligned}$$

Extracting the (i,p) -component with $\vartheta^{i,p} \circ P_{\mathcal{E}_i}$ and using (17),

$$\begin{aligned} P_{\mathcal{E}_i}[\nabla_{P_{\mathcal{E}_j}[v]} \mathbf{grad}(J_i)] &= \sum_{q,p} \left(d(dJ_i[f_{i,p}])[f_{j,q}] \right. \\ &\quad \left. + \sum_{k,\ell} dJ_i[f_{k,\ell}] \vartheta^{i,p}[\nabla_{f_{j,q}} f_{k,\ell}] \right) \vartheta^{j,q}[v] f_{i,p}. \end{aligned} \quad (45)$$

Adding (44) and (45) and canceling identical terms, we get

$$\begin{aligned} A_{ij}[v] &= \sum_{q,p} \left(d(dJ_i[f_{i,p}])[f_{j,q}] \right. \\ &\quad \left. + \sum_{k,\ell} dJ_k[f_{k,\ell}] \vartheta^{i,p}[\nabla_{f_{j,q}} f_{k,\ell}] \right) \vartheta^{j,q}[v] f_{i,p}. \end{aligned} \quad (46)$$

By condition (12) and (19), $\mathbf{grad}_{\mathcal{E}_k}(J_k)(x^*) = 0$ implies

$$dJ_k[f_{k,\ell}]|_{x^*} = 0 \quad \text{for all } k, \ell.$$

Hence the inner sum in (46) vanishes at x^* , leaving

$$A_{ij}[v]|_{x^*} = \sum_{q,p} d(dJ_i[f_{i,p}])[f_{j,q}] \vartheta^{j,q}[v] f_{i,p}|_{x^*}.$$

Reading off the matrix entries with respect to the orthonormal frame gives the first equality in (20). Finally, from the defining identity of the Hessian,

$$\text{Hess}(J_i)[X, Y] = d(dJ_i[Y])[X] - dJ_i[\nabla_X Y],$$

applied with $X = f_{j,q}$ and $Y = f_{i,p}$,

$$d(dJ_i[f_{i,p}])[f_{j,q}] = \mathbf{Hess}(J_i)[f_{i,p}, f_{j,q}] + dJ_i[\nabla_{f_{j,q}} f_{i,p}],$$

establishing the second equality in (20).

D. Proof of Proposition 3

By the first equality in (20),

$$(A_{ii}|_{x^*})_{p,q} = d(dJ_i[f_{i,p}])[f_{i,q}]|_{x^*}.$$

The definition of the Hessian implies that

$$d(dJ_i[f_{i,p}])[f_{i,q}] = \mathbf{Hess}(J_i)[f_{i,q}, f_{i,p}] + dJ_i[\nabla_{f_{i,q}} f_{i,p}]. \quad (47)$$

Since $\mathcal{U}_i$ is a weakly embedded submanifold,

$$\nabla_{f_{i,q}} f_{i,p} = \nabla_{f_{i,q}}^i f_{i,p} + \Pi^i(f_{i,q}, f_{i,p}),$$

where $\nabla_{f_{i,q}}^i f_{i,p} \in \mathcal{E}_i$, and Π^i is the second fundamental form. The equilibrium condition (12) gives $dJ_i[f_{i,r}]|_{x^*} = 0$ for all r , so $dJ_i[\nabla_{f_{i,q}}^i f_{i,p}]|_{x^*} = 0$, and hence $dJ_i[\nabla_{f_{i,q}} f_{i,p}]|_{x^*} = dJ_i[\Pi^i(f_{i,q}, f_{i,p})]|_{x^*}$. From [35, Proposition 1.2], which is obtained by applying the Gauss equation of Riemannian geometry to the Hessian, we obtain that

$$\mathbf{Hess}(J_i^{\mathcal{U}_i})[f_{i,q}, f_{i,p}] = \mathbf{Hess}^i(J_i)[f_{i,q}, f_{i,p}] - dJ_i[\Pi^i(f_{i,q}, f_{i,p})]. \quad (48)$$

Substituting into (47), the second fundamental form contributions cancel, and we obtain

$$d(dJ_i[f_{i,p}])[f_{i,q}]|_{x^*} = \mathbf{Hess}^i(J_i^{\mathcal{U}_i})[f_{i,p}, f_{i,q}]|_{x^*}.$$

E. Proof of Proposition 4

Since $\{f_{i,j}\}$ is a global orthonormal frame on $\mathcal{U}$ with respect to g (see the proof of Theorem 1), we have $g(f_{i,p}, f_{k,r}) = \delta_{ik}\delta_{pr}$, which is constant on $\mathcal{U}$. Therefore, all Lie-derivative terms in the Koszul formula [36, Corollary 5.11] vanish identically, and for any $k \in \{1, \dots, N\}$ and $r \in \{1, \dots, d_k\}$, we have that

$$\begin{aligned} 2g(\nabla_{f_{j,q}} f_{i,p}, f_{k,r}) &= -g(f_{j,q}, [f_{i,p}, f_{k,r}]) \\ &\quad + g(f_{i,p}, [f_{k,r}, f_{j,q}]) \\ &\quad + g(f_{k,r}, [f_{j,q}, f_{i,p}]). \end{aligned} \quad (49)$$

Fixing $i \neq j$, the last term vanishes for all k , since, by Assumption 4, $[f_{j,q}, f_{i,p}] = 0$. Next, we consider each possible value for k :

- For $k \notin \{i, j\}$, Assumption 4 gives that $[f_{i,p}, f_{k,r}] = 0$, $[f_{k,r}, f_{j,q}] = 0$, so the right hand side in (49) vanishes.
- For $k = i$, $[f_{i,p}, f_{k,r}] \in \mathcal{E}_i$ by Assumption 3, so $g(f_{j,q}, [f_{i,p}, f_{i,r}]) = 0$, since $\mathcal{E}_j \perp_g \mathcal{E}_i$ (by construction), and, by Assumption 4, $[f_{i,r}, f_{j,q}] = 0$.
- For $k = j$, Assumption 4 gives that $[f_{i,p}, f_{k,r}] = 0$, whereas, by Assumption 3, we have that $[f_{j,r}, f_{j,q}] \in \mathcal{E}_j$, and, therefore $g(f_{i,p}, [f_{j,r}, f_{j,q}]) = 0$, since $\mathcal{E}_i \perp_g \mathcal{E}_j$ (by construction).

From the above, it follows that $g(\nabla_{f_{j,q}} f_{i,p}, f_{k,r}) = 0$, for all k, r , and, since $\{f_{k,r}\}$ is a global frame, (24) follows.

F. Proof of Proposition 5

First, we fix an order of the global frame as $(f_{1,1}, \dots, f_{1,d_1}, f_{2,1}, \dots, f_{2,d_2})$, and let $n := d_1 + d_2$. From (7b), the linear map sending $f_{i,p} \mapsto \tilde{f}_{i,p}$ has the matrix representation $I + \Gamma^\top$ with respect to this ordered frame, where

$$\Gamma = \begin{bmatrix} 0_{d_1 \times d_1} & \gamma \\ 0_{d_2 \times d_1} & 0_{d_2 \times d_2} \end{bmatrix}, \quad (50)$$

with $\gamma \in \mathbb{R}^{d_1 \times d_2}$ being the matrix of deception parameters γ_j^k . To prove the proposition, it suffices to show that $I + \Gamma^\top$, or, equivalently, $I + \Gamma$, is invertible, since, in that case, $\{\tilde{f}_{i,j}(x)\}$ is a basis of $T_x\mathcal{U}$, the rank- d_i subspaces $\tilde{\mathcal{E}}_i(x)$ are well-defined, and the direct-sum decomposition $T_x\mathcal{U} = \tilde{\mathcal{E}}_1(x) \oplus \tilde{\mathcal{E}}_2(x)$ follows from the basis property. Since the bottom-left block of Γ is zero, the matrix $I + \Gamma$ is block lower-triangular:

$$I + \Gamma = \begin{bmatrix} I_{d_1} & \gamma \\ 0 & I_{d_2} \end{bmatrix},$$

and consequently $\det(I + \Gamma) = \det(I_{d_1}) \cdot \det(I_{d_2}) = 1$, so $I + \Gamma$ is invertible at every $x \in \mathcal{U}$. Since $\{f_{i,j}(x)\}$ is a basis of $T_x\mathcal{U}$ at every $x \in \mathcal{U}$ by Assumption 1, and $I + \Gamma$ is invertible, the tilted family $\{\tilde{f}_{i,j}(x)\}$ is also a basis of $T_x\mathcal{U}$. In particular, $\{\tilde{f}_{i,j}(x)\}_{j=1}^{d_i}$ are linearly independent and span the rank- d_i subspace $\tilde{\mathcal{E}}_i(x)$, and the decomposition $T_x\mathcal{U} = \tilde{\mathcal{E}}_1(x) \oplus \tilde{\mathcal{E}}_2(x)$ follows from dimension counting.

G. Proof of Theorem 3

The proof proceeds in three steps. First, we recast the equilibrium condition (27) as a smooth map $F : \mathcal{U} \times \mathbb{R}^{d_1 \times d_2} \rightarrow \mathbb{R}^n$ whose zero set parametrizes the equilibria of (8) as a function of γ . Second, we apply the implicit function theorem at $(x^*, 0)$ using the non-degeneracy of the nominal Jacobian $A|_{x^*}$ to obtain the smooth map X^* . Third, we use continuity of the spectrum of the deceptive Jacobian $\tilde{A}|_{X^*(\gamma)}$ to transfer local exponential stability from $\gamma = 0$ to a neighborhood Ω .

Let $\{\vartheta^{i,j}\}$ denote the dual co-frame of $\{f_{i,j}\}$ on $\mathcal{U}$, as in (17). The equilibrium condition (27) for system (8) reads

$$\mathbf{grad}_{\tilde{\mathcal{E}}_1}(J_1)(x) = 0, \quad \mathbf{grad}_{\tilde{\mathcal{E}}_2}(J_2)(x) = 0.$$

Since $\tilde{f}_{1,j} = f_{1,j}$, the first condition is independent of γ and amounts to

$$dJ_1[f_{1,j}](x) = 0, \quad j = 1, \dots, d_1.$$

Since $\tilde{f}_{2,j} = f_{2,j} + \sum_{k=1}^{d_1} \gamma_j^k f_{1,k}$, the second condition is

$$dJ_2[\tilde{f}_{2,j}](x) = dJ_2[f_{2,j}](x) + \sum_{k=1}^{d_1} \gamma_j^k dJ_2[f_{1,k}](x) = 0,$$

for $j = 1, \dots, d_2$. We now introduce the smooth map $F : \mathcal{U} \times \mathbb{R}^{d_1 \times d_2} \rightarrow \mathbb{R}^n$ (with $n = d_1 + d_2$) by its n scalar components

$$\begin{aligned} F_{1,j}(x, \gamma) &= dJ_1[f_{1,j}](x), \\ F_{2,j}(x, \gamma) &= dJ_2[f_{2,j}](x) + \sum_{k=1}^{d_1} \gamma_j^k dJ_2[f_{1,k}](x). \end{aligned} \quad (51)$$

Then, $x \in \mathcal{U}$ is an equilibrium of (8) with deception parameter γ if and only if $F(x, \gamma) = 0$. By construction, $F(x^*, 0) = 0$ (the nominal equilibrium conditions (12) at x^*).

The partial derivative $\partial_x F(x^*, 0) : T_{x^*}\mathcal{U} \rightarrow \mathbb{R}^n$ is the linear map with scalar components, for $v \in T_{x^*}\mathcal{U}$,

$$\begin{aligned}\partial_x F_{1,j}(x^*, 0)[v] &= d(dJ_1[f_{1,j}])[v]|_{x^*}, \\ \partial_x F_{2,j}(x^*, 0)[v] &= d(dJ_2[f_{2,j}])[v]|_{x^*}.\end{aligned}$$

Expanding $v = \sum_{i',p'} \vartheta^{i',p'}[v] f_{i',p'}(x^*)$ and using the orthonormality of the frame, we obtain that

$$\partial_x F_{i,j}(x^*, 0)[v] = \sum_{i',p'} d(dJ_i[f_{i,j}])[f_{i',p'}]|_{x^*} \vartheta^{i',p'}[v].$$

By the first equality in (20) of Theorem 2, the coefficients $d(dJ_i[f_{i,j}])[f_{i',p'}]|_{x^*}$ are precisely the entries of the matrix representation of the nominal Jacobian $A|_{x^*}$ with respect to the orthonormal frame $\{f_{i,j}\}$. Therefore, under the identification $\mathbb{R}^n \simeq T_{x^*}\mathcal{U}$ provided by the frame $\{f_{i,j}(x^*)\}$, the linear map $\partial_x F(x^*, 0)$ coincides with the matrix representation of $A|_{x^*}$:

$$\partial_x F(x^*, 0) \simeq A|_{x^*}. \quad (52)$$

By the hypothesis of local exponential stability of x^* and the “only if” direction of Proposition 1, $A|_{x^*}$ is Hurwitz and, in particular, invertible. Hence $\partial_x F(x^*, 0)$ is a linear isomorphism from $T_{x^*}\mathcal{U}$ to $\mathbb{R}^n$.

Next, note that the map F defined in (51) is smooth in both arguments (since $J_1, J_2, f_{i,j}$ are all smooth), $F(x^*, 0) = 0$, and $\partial_x F(x^*, 0)$ is a linear isomorphism. By the implicit function theorem, there exists an open neighborhood $\Omega \subset \mathbb{R}^{d_1 \times d_2}$ of $\gamma = 0$ and a smooth map $X^* : \Omega \rightarrow \mathcal{U}$ such that $X^*(0) = x^*$ and $F(X^*(\gamma), \gamma) = 0$ for all $\gamma \in \Omega$. By construction, $X^*(\gamma)$ is an equilibrium of (8) with deception parameter γ for every $\gamma \in \Omega$, and uniqueness in a neighborhood of x^* is part of the statement of the implicit function theorem.

Finally, let $\tilde{A}(\gamma) := \tilde{A}|_{X^*(\gamma)}$ denote the intrinsic Jacobian of the deceptive dynamics (8), evaluated at the deceptive equilibrium $X^*(\gamma)$. Since both X^* and the frame $\{\tilde{f}_{i,j}\} = \{f_{i,j}(\gamma)\}$ depend smoothly on γ , and since the projected gradients $\text{grad}_{\tilde{\mathcal{E}}_i}(J_i)$ depend smoothly on γ through $\tilde{f}_{i,j}$, the Jacobian $\tilde{A}(\gamma)$ is a smooth function of γ on Ω . At $\gamma = 0$, the deceptive frame $\{\tilde{f}_{i,j}\}$ coincides with the nominal frame $\{f_{i,j}\}$, the deceptive metric $\tilde{g}$ coincides with g , the deceptive bundles $\tilde{\mathcal{E}}_i$ coincide with $\mathcal{E}_i$, and $X^*(0) = x^*$. Consequently,

$$\tilde{A}(0) = A|_{x^*},$$

which is Hurwitz by hypothesis. Since the spectrum of a matrix depends continuously on its entries, and smoothness of $\tilde{A}(\gamma)$ in γ implies continuity of its entries, there exists an open neighborhood $\Omega' \subset \Omega$ of $\gamma = 0$ such that $\tilde{A}(\gamma)$ remains Hurwitz for all $\gamma \in \Omega'$. Shrinking Ω to Ω' if necessary, the “if” direction of the second statement of Proposition 1 applied to (8) gives that $X^*(\gamma)$ is a locally exponentially stable equilibrium of (8) for all $\gamma \in \Omega$.

H. Proof of Corollary 3

Differentiating the identity $F(X^*(\gamma), \gamma) = 0$ with respect to γ_j^k at $\gamma = 0$ gives

$$\partial_x F(x^*, 0) \left[\frac{\partial X^*}{\partial \gamma_j^k} \Big|_{\gamma=0} \right] + \frac{\partial F}{\partial \gamma_j^k}(x^*, 0) = 0.$$

From (51), the only component of F that depends on γ_j^k is $F_{2,j}$, and $\partial F_{2,j} / \partial \gamma_j^k(x^*, 0) = dJ_2[f_{1,k}](x^*)$. Under the identification (52), the vector $\partial F / \partial \gamma_j^k(x^*, 0) \in \mathbb{R}^n$ corresponds to $dJ_2[f_{1,k}](x^*) f_{2,j}(x^*) \in T_{x^*}\mathcal{U}$, and inverting $\partial_x F(x^*, 0) \simeq A|_{x^*}$, which is invertible by assumption, yields (28).

VII. CONCLUSION

We studied model-free multi-agent decision making on manifolds under payoff-only measurements and control-affine actuation along state-dependent vector fields. Using a Lie-bracket approximation viewpoint, we showed that, under a natural bundle-splitting condition, the interaction dynamics admits an intrinsic geometric representation as a sum of projected Riemannian gradients, i.e., as a non-holonomic gradient play, which yields an equilibrium characterization and an intrinsic local stability test via a connection-based linearization. We then introduced an asymmetric-information model suitable for manifold-constrained interactions and showed how embedding victims’ exploration signals into the strategy of a deceiver tilts the victims’ effective actuation directions, generically shifting equilibria and enabling the deceiver to control the location of the equilibria, with an explicit first-order characterization of the induced shift.

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REFERENCES

- [1] A. Sarlette, R. Sepulchre, and N. E. Leonard, “Autonomous rigid body attitude synchronization,” *Automatica*, vol. 45, no. 2, pp. 572–577, 2009.
- [2] W. Ren, “Distributed cooperative attitude synchronization and tracking for multiple rigid bodies,” *IEEE Transactions on Control Systems Technology*, vol. 18, no. 2, pp. 383–392, 2009.
- [3] J. Poveda and N. Quijano, “Distributed extremum seeking for real-time resource allocation,” in *2013 American Control Conference*, pp. 2772–2777, IEEE, 2013.
- [4] D. E. Ochoa, J. I. Poveda, C. A. Uribe, and N. Quijano, “Hybrid robust optimal resource allocation with momentum,” in *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 3954–3959, IEEE, 2019.
- [5] H. Yin, P. G. Mehta, S. P. Meyn, and U. V. Shanbhag, “Synchronization of coupled oscillators is a game,” *IEEE Transactions on Automatic Control*, vol. 57, no. 4, pp. 920–935, 2011.
- [6] J. S. Shamma and G. Arslan, “Dynamic fictitious play, dynamic gradient play, and distributed convergence to nash equilibria,” *IEEE Transactions on Automatic Control*, vol. 50, no. 3, pp. 312–327, 2005.
- [7] S. A. Toonsi and J. S. Shamma, “Higher-order gradient play leading to nash equilibrium in the bandit setting,” in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pp. 5984–5989, IEEE, 2024.
- [8] L. J. Ratliff, S. A. Burden, and S. S. Sastry, “On the characterization of local Nash equilibria in continuous games,” *IEEE transactions on automatic control*, vol. 61, no. 8, pp. 2301–2307, 2016.
- [9] E. Mazumdar, L. J. Ratliff, and S. S. Sastry, “On gradient-based learning in continuous games,” *SIAM Journal on Mathematics of Data Science*, vol. 2, no. 1, pp. 103–131, 2020.
- [10] P. Frihauf, M. Krstic, and T. Basar, “Nash equilibrium seeking in non-cooperative games,” *IEEE Transactions on Automatic Control*, vol. 57, no. 5, pp. 1192–1207, 2011.
- [11] S.-J. Liu and M. Krstić, “Stochastic nash equilibrium seeking for games with general nonlinear payoffs,” *SIAM Journal on Control and Optimization*, vol. 49, no. 4, pp. 1659–1679, 2011.

- [12] M. S. Stankovic, K. H. Johansson, and D. M. Stipanovic, "Distributed seeking of nash equilibria with applications to mobile sensor networks," *IEEE Transactions on Automatic Control*, vol. 57, no. 4, pp. 904–919, 2011.
- [13] S. Krilasevic and S. Grammatico, "Learning generalized nash equilibria in multi-agent dynamical systems via extremum seeking control," *Automatica*, vol. 133, p. 109846, 2021.
- [14] F. Taringoo, "Synchronization on lie groups: Coordination of blind agents," *IEEE Transactions on Automatic Control*, vol. 62, no. 12, pp. 6324–6338, 2017.
- [15] H.-B. Dürr, M. S. Stanković, K. H. Johansson, and C. Ebenbauer, "Extremum seeking on submanifolds in the euclidian space," *Automatica*, vol. 50, no. 10, pp. 2591–2596, 2014.
- [16] M. Abdelgalil and J. I. Poveda, "On lie-bracket averaging for hybrid dynamical systems with applications to model-free control and optimization," *IEEE Transactions on Automatic Control*, vol. 70, no. 7, pp. 4655–4670, 2025.
- [17] F. Taringoo, P. M. Dower, D. Nesic, and Y. Tan, "Optimization methods on riemannian manifolds via extremum seeking algorithms," *SIAM Journal on Control and Optimization*, vol. 56, no. 5, pp. 3867–3892, 2018.
- [18] D. E. Ochoa and J. I. Poveda, "Robust global optimization on smooth compact manifolds via hybrid gradient-free dynamics," *Automatica*, vol. 171, p. 111916, 2025.
- [19] D. Monderer and L. S. Shapley, "Fictitious play property for games with identical interests," *Journal of economic theory*, vol. 68, no. 1, pp. 258–265, 1996.
- [20] M. Tang, M. Krstic, and J. Poveda, "Deception in oligopoly games via adaptive nash seeking systems," in *International Conference on Game Theory for Networks*, pp. 61–74, Springer, 2025.
- [21] M. Tang, U. Javed, X. Chen, M. Krstic, and J. I. Poveda, "Deception in nash equilibrium seeking," *IEEE Transactions on Automatic Control*, 2025.
- [22] M. Tang, M. Krstic, and J. I. Poveda, "Stochastic real-time deception in nash equilibrium seeking for games with quadratic payoffs," in *Proceedings of the 7th Annual Learning for Dynamics & Control Conference*, Proceedings of Machine Learning Research, 2025.
- [23] M. Abdelgalil, J. I. Poveda, V. Grushkovskaya, A. Zuyev, R. Suttner, and C. Ebenbauer, "Geometric methods in extremum seeking control," *IEEE Conference on Decision and Control*, to appear, 2026.
- [24] K. G. Vamvoudakis, J. I. Poveda, S. Anderson, J. P. Hespanha, M. Krstic, I. R. Petersen, M. Tang, A. Wadi, and Q. Zhu, "A tutorial for foundations in ai and quantum systems: From control and game-theory to stochastic deception," in *2026 American Control Conference (ACC)*, pp. 3300–3317, IEEE, 2026.
- [25] H.-B. Dürr, M. S. Stanković, C. Ebenbauer, and K. H. Johansson, "Lie bracket approximation of extremum seeking systems," *Automatica*, vol. 49, no. 6, pp. 1538–1552, 2013.
- [26] V. Grushkovskaya, A. Zuyev, and C. Ebenbauer, "On a class of generating vector fields for the extremum seeking problem: Lie bracket approximation and stability properties," *Automatica*, vol. 94, pp. 151–160, 2018.
- [27] M. Abdelgalil, D. E. Ochoa, and J. I. Poveda, "Multi-time scale control and optimization via averaging and singular perturbation theory: From odes to hybrid dynamical systems," *Annual Reviews in Control*, vol. 56, p. 100926, 2023.
- [28] M. Abdelgalil, M. Krstic, and J. I. Poveda, "Stochastic lie-bracket approximations for zeroth-order optimization on manifolds," *submitted to IEEE Transactions on Automatic Control*, 2026.
- [29] J. M. Lee in *Introduction to Smooth Manifolds*, Springer, 2012.
- [30] A. Scheinker, "Model independent beam tuning," in *Proceedings of the 2013 International Particle Accelerator Conference, Shanghai, China*, pp. 12–17, 2013.
- [31] A. Scheinker and M. Krstić, "Extremum seeking with bounded update rates," *Systems & Control Letters*, vol. 63, pp. 25–31, 2014.
- [32] L. Moreau and D. Aeyels, "Practical stability and stabilization," *IEEE Transactions on Automatic Control*, vol. 45, no. 8, pp. 1554–1558, 2002.
- [33] K. G. Vamvoudakis, F. Fotiadis, T. Başar, V. Gupta, J. Poveda, M. Tang, M. Krstic, and Q. Zhu, "Deception in game theory and control: A tutorial," in *2025 American Control Conference (ACC)*, pp. 2146–2163, IEEE, 2025.
- [34] D. G. Feingold and R. S. Varga, "Block diagonally dominant matrices and generalizations of the gerschgorin circle theorem," 1962.
- [35] M. Dajczer and R. Tojeiro, *Submanifold Theory: Beyond an Introduction*. Springer, 2019.
- [36] J. M. Lee, *Introduction to Riemannian Manifolds*. Springer, 2018.



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